Unit - I
INTRODUCTION

What is Green Building, Why to go for Green Building, Benefits of Green Buildings, Green Building Materials and Equipment in India, What are key Requisites for Constructing a Green Building, Important Sustainable features for Green Building,

1.1 Green Building

A 'green' building is a building that, in its design, construction or operation, reduces or eliminates negative impacts, and can create positive impacts, on our climate and natural environment. Green buildings preserve precious natural resources and improve our quality of life.

Sustainable Development

- "Sustainable development is development that meets the needs of the present, without compromising the ability of future generations to meet their own needs."
- Sustainable development is about finding better ways of doing things, both for the future and the present.

1.2 Why to go for Green Buildings

The global construction and building sector is one of the largest contributors to climate change and resource depletion. Continuing down the traditional path poses severe risks to our economies, health, and climate stability.

1. The Climate Crisis & Carbon Budgets

The building sector is responsible for nearly 39% of global energy-related carbon emissions. This is split into two categories:

- Operational Emissions (28%): The energy used to heat, cool, and light buildings daily.
- Embodied Carbon (11%): The carbon footprint of manufacturing, transporting, and assembling materials like concrete and steel.

We cannot hit global net-zero climate targets without decarbonizing buildings. Green buildings are a necessity to keep global temperature increases within safe limits and prevent catastrophic climate events.

2. Escalating Resource Scarcity

Traditional construction is incredibly resource-intensive, straining finite global supplies.

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- Water Stress: Global water scarcity is rising rapidly. Traditional buildings put massive pressure on municipal water supplies. Green buildings use rainwater harvesting and greywater recycling to significantly cut demand.
- Material Depletion**:** Construction consumes billions of tons of raw materials annually. Green buildings demand the use of recycled, rapidly renewable, and circular materials, preventing the total depletion of natural resources.

3. Skyrocketing Energy Costs & Grid Failures

The global energy grid is under immense stress due to rising temperatures and increased cooling demands.

- Traditional, poorly insulated buildings act like energy sieves, putting massive strain on electrical grids during peak summer and winter months.
- Green buildings utilize passive design—using smart insulation, strategic window placement, and natural ventilation—to stay comfortable without relying heavily on the grid. They reduce the risk of rolling blackouts and protect occupants from skyrocketing energy prices.

4. Regulatory and Policy Mandates

Governments worldwide are rapidly shifting green regulations from "encouraged" to "legally mandated."

- Many major global cities are enforcing strict carbon limits on large buildings, introducing heavy fines for properties that exceed emission caps.
- Financial institutions are increasingly tying commercial loans and investment capital to ESG (Environmental, Social, and Governance) scores.

Soon, traditional buildings will become stranded assets—unsellable, unrentable, and illegal to operate under modern environmental laws. Building green is now a necessity for basic financial survival in the real estate market.

5. Urban Heat Islands & Public Health

As cities expand with concrete and asphalt, they trap heat, creating "urban heat islands" that make cities significantly hotter than surrounding rural areas.

- Green buildings counter this by integrating green roofs (vegetation), living walls, and cool, reflective pavements. Without them, our growing urban centres' will face unbearable, dangerous heat levels during summer months.

1.3 Benefits of Green Building

Going for Green Buildings have tremendous benefits, both tangible and intangible.

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The most tangible benefit is in reduction of operating energy and water costs right from day one during the entire life cycle of the building

Green Buildings offer a range of benefits like:

- 30% to 40% reduction in operation cost
- Green Corporate image
- Health and safety of building occupants
- Enhance occupant comfort
- Improve productivity of occupants
- Imbibe best operational practices from day one
- Incorporates latest techniques and technologies

Environmental Benefits

- Reduces impact of natural resource consumption
- Develops long-term resource/energy strategies that are practical in use

Economic and Social Benefits

- Reduces reliance on public utilities
- Increases the value of a home and increases resale interest
- Increases home durability
- Encourages sustainable building practices
- Reduce the impact on natural resource consumption and leave for future generations

Health and Safety Benefits

- Reduces exposure to allergens, carcinogens, pollutants
- Enhances occupant comfort and health
- Improve productivity of occupants

Community Benefits

- Reduces strain on local infrastructures and improves quality of life
- Provides homeowner pride and drives values for community members

Competitive Costs

- Integrated Green Design allows higher benefits at lower cost by achieving synergies between disciplines and between technologies

Operating and Tax Costs

- Utility costs significantly lower
- Life-cycle economic performance optimized
- Owner access to tax credits

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Increases Building Valuation and ROI

- Quantifying financial benefit in terms of return on investment (ROI) instead of payback time demonstrates true financial advantage

Faster environmental clearances for projects

- In a major impetus to the Green Buildings movement in India, Government of Maharashtra has in-principle approved Priority Status for "Faster environmental Clearance for IGBC registered projects" from the Environment Department and other Municipalities.

1.4 Green Building Materials and Equipment in India

1.4.1 Green Building Materials

i. Fly-ash bricks & Fly-ash based blocks

- Manufactured using coal-fired power plant waste (fly ash). Lighter, uniform, good compressive strength, lower water absorption and reduced top-soil depletion vs traditional clay bricks. Widely adopted in India for walls, pavements.

ii. Pozzolana Cement

- Portland Pozzolana cement is integrated cement which is formed by synthesising OPC cement with pozzolanic materials in a certain proportion. It is commonly known as PPC cement.
- A pozzolanic material can be a natural or artificial which contains silica and aluminous in a reactive form. These materials usually don't possess any cementitious properties, but when it is mixed with water or moisture or lime to undergo reaction with calcium hydroxide to form compounds possessing cement properties.

iii. AAC (Autoclaved Aerated Concrete) blocks

- Lightweight, insulating precast blocks — reduce wall weight and cooling loads, speed construction. Popular in mid-to-high rise and commercial projects.

iv. Compressed Stabilized Earth Blocks (CSEB) & Stabilized Mud Blocks

- Locally-made earth mixed with stabilizer (cement/lime) and compressed. Low embodied energy, good thermal mass, low cost where suitable soils exist. Good for vernacular and low-rise construction.

v. Bamboo

- Bamboo as a building material is used for the construction of scaffolding, bridges and structures, houses.
- Bamboos are one of the fastest-growing plants in the world and their growth is three times faster than most other species of plants.
- They are renewable and extremely versatile resource with multi-purpose usage.

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- Among many uses of bamboo, Housing is one of the major areas applications
- especially in the wake of residential shortages around the globe.

vi. Recycled / Secondary Steel & Rebar with high recycled content

- Steel with high recycled content reduces embodied energy vs primary steel. Use from reputable manufacturers and check mill/test certificates.

vii. Low-VOC paints, adhesives & finishes

- Low volatile organic compound products reduce indoor air pollution and occupant health risks. Many manufacturers in India offer low/no-VOC lines.

viii. Insulation materials (mineral wool, EPS, PIR, sheep-wool alternatives)

- Proper insulation for walls/roof reduces heating/cooling loads. Choose fire-rated and moisture-resistant options as per application.

ix. High-performance glazing & fenestration

- Double glazing, low-e coatings, proper frame thermal breaks reduce heat gain and improve daylighting control.

x. Green roof / vegetated roof systems and cool roofing membranes

- Reduce urban heat island effect, add insulation, manage stormwater. Cool reflective roof coatings are widely used when green roofs aren't practical.

xi. Permeable paving & recycled aggregates

- Permeable pavers reduce runoff; crushed/recycled concrete aggregates lower demand for virgin aggregates.

Bamboo:

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Bamboo can be shaped in the following ways

Squared cross-section can be obtained by compressing the growing stalk of bamboo within a square section.

Arch shapes of bamboo can also be created by compressing the bamboo's growth into the desired shape. This would cost lesser than it would to get the same form with normal timber.

Curved and Flat shapes of bamboo are achieved through traditional techniques like applying heat and pressure.

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The various advantages of bamboo are as mentioned below:

- Tensile strength: Bamboo has higher tensile strength than steel because its fibers run axially.
- Fire Resistance: Capability of bamboo to resist fire is very high and it can withstand temperature up to 400^0 C. This is due to the presence of high value of silicate acid and water.
- Elasticity: Bamboo is widely preferred in earthquake prone regions due to its elastic features.
- Weight of bamboo: Bamboos due to their low weight are easily displaced or installed making it very easier for transportation and construction.
- Unlike other building materials like cement and asbestos, bamboo poses no danger to health.
- They are cost effective and easy to use.
- They are especially in great demand in earthquake prone areas.

Eucalyptus:

Eucalyptus wood, a fast-growing hardwood, is sustainable because new growth only takes a few years to mature.

Building with eucalyptus wood is desirable because, unlike other timbers, it has very little taper over long lengths. This quality makes it more consistent and efficient. There are more timber and less waste. And generally speaking (each Eucalyptus species is slightly different) it is harder that most North American woods.

Because eucalyptus wood is sustainable, it is now being used as an alternative to mahogany and Brazilian cherry. Both of these species are endangered, slow-growing and comparatively expensive. Building with eucalyptus wood is quickly becoming a natural choice.

Pressure treated eucalyptus poles have at least a 20-year ground burial rate.

Poplar Wood:

Wood from the yellow poplar tree is often used in light construction, such as in single-family homes or storage sheds. Wood from the yellow poplar tree is also used in kitchen cabinets and doors. Since wood from the yellow poplar tree can be easily machine-worked, painted and stained, this wood is an excellent choice for furniture making as well.

Rubberwood:

- Both eco-friendly and cost effective, rubber wood is ideal for furniture, flooring, ceiling, paneling and interiors
- Treated rubber wood possess advantages like
- Excellent stability
- Excellent machining properties
- Good strength properties
- Light color; easy to finish in any color

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- Good glaring properties
- Attractive grain structure
- Eco-friendly
- Up to 40% cheaper than other hardwoods

Linoleum:

Linoleum, also called Lino, is a floor covering made from materials such as solidified linseed oil (linoxyn), pine rosin, ground cork dust, wood flour, and mineral fillers such as calcium carbonate, most commonly on a burlap or canvas backing. Pigments are often added to the materials to create the desired colour finish.

Rammed Earth:

Rammed earth walls are constructed by ramming a mixture of selected aggregates, including gravel, sand, silt and a small amount of clay, into place between flat panels called formwork.

Traditional technology repeatedly rammed the end of a wooden pole into the earth mixture to compress it. Modern technology replaces the pole with a mechanical ram.

The many advantages of building with rammed earth include superior thermal mass, temperature and noise control, strength and durability, low maintenance, fire proofing, load bearing and pest deterrence, as well as its beauty and the pleasure of building with a natural and environmentally sound material.

Stabilized Mud:

The input of soil stabilization allowed people to build higher with thinner walls, which have a much better compressive strength and water resistance. With cement stabilization, the blocks must be cured for four weeks after manufacturing. After this, they can dry freely and be used like common bricks with a soil cement stabilized mortar.

Ideally, the production is made on the site itself or in the nearby area. Thus, it will save the transportation, fuel, time and money.

Well-designed Compressed Stabilized Earth Blocks (CSEB) houses can withstand, with a minimum of maintenance, heavy rains, snowfall or frost without being damaged. The strength and durability have been proven since half a century.

Adobe bricks:

Adobe bricks (mud bricks) are made of earth with a fairly high clay content and straw. The produced earth mix is cast in open moulds onto the ground and then left to dry out. Adobe bricks are only sun-dried, not kiln-fired. When used for construction they are laid up into a wall using an earth mortar. Before drying out, the finished walls are smoothed down. Often a clay render is applied as a surface coating.

As with other forms of earth construction, adobe bricks are a fireproof, durable yet biodegradable, non-toxic building material which provides sufficient thermal mass to buildings

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to ensure excellent thermal performance. Other benefits include low sound transmission levels through walls and a general feeling of solidity and security.

1.4.2 Green Building Equipment in India

i. *Solar Photovoltaic (rooftop) Systems*- Onsite renewable energy systems, particularly rooftop solar photovoltaic installations, help reduce dependence on the electricity grid, lower a building's carbon footprint, and contribute to sustainable energy generation. In many Indian cities, these systems offer attractive payback periods due to rising electricity tariffs and abundant solar resources. Furthermore, the Ministry of New and Renewable Energy (MNRE), Government of India, along with various state governments, supports rooftop solar adoption through incentives, subsidies, and net-metering policies, enabling consumers to export surplus electricity to the grid and receive credits against their energy consumption.

ii. *Solar Water Heaters (SWH)*- Solar water heating is a mature and widely adopted technology in India for meeting domestic, commercial, and institutional hot-water requirements. It significantly reduces electricity consumption by replacing or minimizing the use of conventional electric geysers, resulting in lower energy costs and reduced greenhouse gas emissions. The technology is reliable, cost-effective, and well-suited to India's climatic conditions, with numerous MNRE-listed manufacturers and vendors providing certified systems, installation, and maintenance services across the country.

iii. *Efficient HVAC: VRF/VRV, Heat Pumps, High-SEER Equipment*- HVAC systems are often the largest consumers of energy in commercial buildings; therefore, specifying high-efficiency equipment, Variable Refrigerant Flow (VRF) systems, and heat-recovery technologies wherever applicable can significantly reduce energy consumption and operating costs. The incorporation of thermostatic controls, programmable thermostats, and zoning strategies further enhances efficiency by enabling precise temperature regulation and conditioning only occupied spaces, thereby improving occupant comfort while minimizing unnecessary energy use.

iv. *LED Lighting & Smart Lighting Controls (occupancy sensors, daylight dimming)*- LED lighting systems can reduce lighting energy consumption by approximately 60–80% compared to traditional incandescent lamps and offer significant energy savings over older fluorescent lighting technologies. In addition to their higher efficiency, LEDs provide longer service life, lower maintenance requirements, and improved lighting quality. Further energy savings can be achieved by integrating occupancy sensors, motion detectors, daylight harvesting controls, and automatic dimming systems, which ensure that lighting is used only when and where it is needed while maximizing the utilization of available natural daylight.

v. *Building Management System (BMS) / Building Automation System (BAS)*- It integrates and centrally control building services such as HVAC, lighting, pumps, elevators, and renewable energy systems to optimize overall building performance. These systems continuously monitor energy consumption, equipment operation, and environmental conditions, enabling automated control strategies that improve efficiency, occupant comfort, and maintenance management. By providing real-time data, fault detection,

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and energy analytics, BMS/BAS solutions help reduce operational costs and energy wastage, making them particularly valuable and often essential for larger commercial, institutional, and industrial projects seeking to maximize long-term operational savings and sustainability performance.

vi. *Variable Frequency Drives (VFDs) for pumps & fans*- Variable Frequency Drives (VFDs) for Pumps and Fans enable motor speeds to be adjusted according to actual system demand rather than operating continuously at full speed. By matching the output of pumps and fans to varying load requirements, VFDs significantly reduce energy consumption, improve process control, and minimize mechanical wear and maintenance costs. They are widely recognized as a standard energy-efficiency measure for pumping systems, cooling towers, and central air-handling unit (AHU) fans, offering substantial operational savings while enhancing overall system performance and reliability.

vii. *Rainwater harvesting (RWH) & Groundwater recharge systems*- Rainwater Harvesting (RWH) is a critical water conservation measure, particularly in water-stressed Indian cities where increasing urbanization and groundwater depletion have intensified water scarcity challenges. Many state governments and urban local bodies have made RWH mandatory through municipal building regulations and development bylaws. Rainwater harvesting systems can range from simple groundwater recharge pits and percolation wells to advanced rooftop collection, storage, filtration, and reuse systems integrated into building infrastructure. These systems help reduce dependence on municipal water supplies, enhance groundwater recharge, and improve water security. To ensure compliance and effectiveness, several cities have initiated inspections and enforcement actions, including monitoring non-functional or poorly maintained RWH installations and implementing penalties where mandated systems are not operational.

viii. *Sewage Treatment Plants (STP) / Decentralized wastewater treatment & Greywater recycling*- Sewage Treatment Plants (STPs), decentralized wastewater treatment systems, and greywater recycling systems play a vital role in sustainable water management by enabling the treatment and reuse of wastewater within a building or campus. Treated wastewater can be effectively reused for non-potable applications such as toilet flushing, landscape irrigation, cooling tower makeup, and cleaning purposes, significantly reducing the demand for fresh potable water. Decentralized treatment systems are particularly suitable for residential complexes, institutional campuses, commercial buildings, and townships, where wastewater can be managed close to its source. The adoption of greywater recycling and wastewater reuse not only conserves water resources but also reduces the burden on municipal sewerage infrastructure. In India, several national and state-level manuals, standards, and guidelines have been developed to support the planning, design, operation, and maintenance of greywater management and wastewater reuse systems, promoting their integration into sustainable and green building practices.

ix. *Efficient pumps, water-efficient fixtures (low flow taps, dual-flush Water Closets)*- Efficient pumps and water-efficient fixtures, such as low-flow taps, aerators, sensor-based faucets, low-flow showerheads, and dual-flush water closets (WCs), are effective measures for reducing overall water consumption in buildings. By minimizing water

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demand, these systems also reduce the energy required for water pumping, treatment, and distribution, thereby contributing to both water and energy conservation. High-efficiency pumps further optimize system performance by delivering the required flow with lower power consumption. The Bureau of Indian Standards (BIS), along with green building rating systems such as IGBC and GRIHA, specifies performance criteria and benchmarks for water-efficient fixtures and pumping systems, encouraging their adoption as part of sustainable building design and operation.

x. *Energy-efficient elevators and regenerative drives*- Energy-efficient elevators and regenerative drive systems contribute to reducing the overall energy consumption of buildings by optimizing elevator operation and recovering energy that would otherwise be wasted. Regenerative drives capture the electrical energy generated when an elevator car travels downward with a heavy load or upward with a light load and feed this energy back into the building's electrical system for use by other equipment. In addition to regenerative technology, modern energy-efficient elevators incorporate features such as variable voltage variable frequency (VVVF) drives, LED cabin lighting, automatic standby modes, and intelligent dispatch controls, which further enhance energy performance. These systems are particularly beneficial in high-rise residential, commercial, and institutional buildings, where elevator usage is significant and energy savings can be substantial over the building's lifecycle.

1.5 Key Requisites for Constructing a Green Building

The key requisites for constructing a *Green Building* are the planning, design, construction, and operation practices that minimize environmental impact while ensuring occupant comfort and resource efficiency.

1. Sustainable Site Selection and Planning

- Choose sites with minimal ecological disturbance.
- Avoid construction on wetlands, forests, and fertile agricultural land.
- Ensure easy access to public transportation.
- Preserve existing vegetation and natural drainage patterns.
- Reduce heat island effect through landscaping and reflective surfaces.

Criteria for Site Selection

Do not develop buildings on portions of sites that meet any of the following on a site

a. Prime farmland
b. Previously under-developed land whose elevation is lower than 1.5m above the elevation of the 100yrs flood frequency.
c. Land specifically identified as habitat for any species on tranced or endangered list.
d. Land within 30m of any wetland and isolated wetlands or areas of special concern identified state or central rule
e. Previously under developed land that is within 20m of any water-body such as lakes, rivers and reservoirs.

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f. Site shall not be located near to solid waste disposal sites, oxidation ponds and odour producing industries.
g. The site location shall be close to suburban transportation route, and is within walking, biking, or bus distance to medical clinics, schools, parks, temples, shopping, parks and other recreational spots.
h. The site location should demonstrate cultural diversity consistent with that statewide.
i. The site shall not be located on brownfield which is a place of contamination. With hazardous substance or pollutants
j. The site location should not spoil the existing land scape and protect it from degradation during the process of construction.
k. The site soil should be good in terms of soil bearing capacity, compatibility and soil permeability

2. Energy-Efficient Building Design

- Orient the building to maximize natural lighting and ventilation.
- Use passive solar design principles.
- Optimize building envelope (walls, roofs, windows) for thermal performance.
- Install high-performance glazing and insulation.
- Incorporate energy-efficient HVAC systems.

3. Renewable Energy Integration

- Install solar photovoltaic (PV) panels.
- Use solar water heating systems.
- Consider wind energy or other renewable sources where feasible.
- Implement battery storage systems for improved energy management.

4. Water Conservation Measures

- Rainwater harvesting systems.
- Low-flow fixtures and water-efficient appliances.
- Greywater recycling systems.
- Efficient irrigation methods such as drip irrigation.
- Water metering and leak detection systems.

5. Sustainable Building Materials

- Use locally sourced materials to reduce transportation emissions.
- Select recycled and recyclable materials.
- Employ rapidly renewable materials such as bamboo.
- Use low embodied-energy materials.
- Avoid toxic and high-VOC (Volatile Organic Compound) materials.

6. Efficient Waste Management

- Segregate construction waste at source.

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- Reuse and recycle demolition and construction debris.
- Provide facilities for occupant waste segregation.
- Minimize landfill disposal.

7. Indoor Environmental Quality (IEQ)

- Ensure adequate fresh air ventilation.
- Maximize daylight penetration.
- Control indoor pollutants.
- Use low-VOC paints, adhesives, and finishes.
- Maintain thermal, visual, and acoustic comfort.

8. Green Landscaping

- Plant native and drought-resistant species.
- Develop green roofs and vertical gardens where feasible.
- Create shaded outdoor spaces.
- Enhance biodiversity around the site.

9. Smart Building Management

- Install Building Management Systems (BMS).
- Use occupancy sensors and automated lighting controls.
- Monitor energy and water consumption continuously.
- Implement predictive maintenance strategies.

10. Climate-Responsive Design

- Design according to local climate conditions.
- Utilize natural cooling and heating techniques.
- Provide protection against extreme weather events.
- Improve resilience to climate change impacts.

11. Life-Cycle Cost Consideration

- Evaluate materials and systems based on life-cycle cost rather than initial cost.
- Consider operation, maintenance, replacement, and disposal costs.
- Aim for long-term economic and environmental benefits.

12. Compliance with Green Building Rating Systems

Common certification systems include:

- Indian Green Building Council (IGBC)
- Green Rating for Integrated Habitat Assessment (GRIHA)
- Leadership in Energy and Environmental Design (LEED)

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1.6 Important Sustainable features for Green Building

These include

- Efficient use of energy, water and other resources
- Use of renewable energy, such as solar energy
- Pollution and waste reduction measures, and the enabling of re-use and recycling
- Good indoor environmental air quality
- Use of materials that are non-toxic and sustainable
- Consideration of the environment in design, construction and operation
- Consideration of the quality of life of occupants in design, construction and operation
- A design that enables adaptation to a changing environment

Energy Efficiency

- Passive design strategies can dramatically affect building energy performance.
- Develop strategies to provide natural lighting.
- Install high-efficiency lighting systems with advanced lighting controls.
- Use a properly sized and energy-efficient heat/cooling system in conjunction with a thermally efficient building shell.
- Minimize the electric loads from lighting, equipment, and appliances
- Consider alternative energy sources such as photo voltaic and fuel cells

Materials Efficiency

- Select sustainable construction materials and products by evaluating several characteristics
- Use dimensional planning and other material efficiency strategies.
- Reuse and recycle construction and demolition materials.
- Require plans for managing materials through deconstruction, demolition, and construction.
- Design with adequate space to facilitate recycling collection and to incorporate a solid waste management program that prevents waste generation.

Water Efficiency

- Design for dual plumbing to use recycled water for toilet flushing or a gray water system that recovers rainwater or other non-potable water.
- Minimize wastewater by using ultra-low-flush toilets, low-flow shower heads, and other water conserving fixtures.
- Use recirculating systems for centralized hot water distribution.
- Install point-of-use hot water heating systems for more distant locations.
- Use state-of-the-art irrigation controllers and self-closing nozzles on hoses.

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Occupant Health and Safety

- Choose construction materials and interior finish products with zero or low emissions to improve indoor air quality.
- Many building materials and cleaning/maintenance products emit toxic gases, such as volatile organic compounds (VOC) and formaldehyde. These gases can have a detrimental impact on occupants' health and productivity.
- Provide adequate ventilation and a high-efficiency, in-duct filtration system.
- Prevent indoor microbial contamination through selection of materials resistant to microbial growth, provide effective drainage from the roof and surrounding landscape, install adequate ventilation in bathrooms, allow proper drainage of air-conditioning coils, and design other building systems to control humidity.

Building Operation and Maintenance

- Green building measures cannot achieve their goals unless they work as intended.
- Over time, building performance can be assured through measurement, adjustment, and upgrading.
- Proper maintenance ensures that a building continues to perform as designed and commissioned.

Any building can be a green building, whether it’s a home, an office, a school, a hospital, a community centre, or any other type of structure, provided it includes features listed above. However, it is worth noting that not all green buildings are – and need to be - the same.

Different countries and regions have a variety of characteristics such as distinctive climatic conditions, unique cultures and traditions, diverse building types and ages, or wide-ranging environmental, economic and social priorities – all of which shape their approach to green building.

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UNIT – II

Green Building Concepts and Practices

Indian Green Building Council, Green Building Moment in India, Benefits Experienced in Green Buildings, Launch of Green Building Rating Systems, Residential Sector, Market Transformation; Green Building Opportunities and Benefits: Opportunities of Green Building, Green Building Features, Material and Resources, Water Efficiency, Optimum Energy Efficiency, Typical Energy Saving Approach in Buildings, LEED India Rating System and Energy Efficiency

2.1 Indian Green Building Council (IGBC)

2.1.1 Green Building Moment in India

The housing sector in India is growing at a rapid pace and contributing immensely to the growth of the economy. This augurs well for the country and now there is an imminent need to introduce green concepts and techniques in this sector, which can aid growth in a sustainable manner.

Green concepts and techniques in the residential sector can help address national issues like water efficiency, energy efficiency, and reduction in fossil fuel use in commuting, handling of consumer waste and conserving natural resources. Most importantly, these concepts can enhance occupant health, happiness and well-being.

Against this background, the Indian Green Building Council (IGBC) has launched 'IGBC Green Homes® Rating System' to address the National priorities. By applying IGBC Green Homes® criteria, homes which are sustainable over the life cycle of the building can be constructed. This rating programme is a tool which enables the designer to apply green concepts and criteria, so as to reduce the environmental impacts, which are measurable. The programme covers methodologies to cover diverse climatic zones and changing lifestyles.

IGBC Green Homes® is the first rating programme developed in India, exclusively for the residential sector. It is based on accepted energy and environmental principles and strikes a balance between known established practices and emerging concepts. The system is designed to be comprehensive in scope, yet simple in operation.

2.1.2 Benefits of Green Homes

Green homes can have tremendous benefits, both tangible and intangible.

1. The most tangible benefits are
    - the reduction in water and energy consumption right from day one of occupancy.
    - the energy savings could range from 20 - 30 % and water savings around 30 - 50%.
2. Intangible benefits of green homes include:
    - enhanced air quality,
    - excellent daylighting,

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- health & well-being of the residents,
- safety benefits and
- conservation of scarce national resources.

2.1.3 Launch of Green Building Rating Systems- Residential Sector

2.1.3.1 Indian Green Building Council (IGBC)

IGBC has set up the Green Homes Core Committee to focus on residential sector. The committee includes builders, developers, home owners, architects, consultants, experts on building science, manufacturers and industry representatives. The varied experience and professions of the committee members bring in a holistic perspective in the process of developing the rating programme.

Evolution of Rating System-IGBC

IGBC, in its endeavour to extend green building concepts to all building types envisioned a rating programme exclusively for homes.

- A core committee was formed under the leadership of Ar Sharukh Mistry, Mistry Architects, Bangalore.
- The pilot version of the programme was launched in July 2008. The rating programme has evoked tremendous response from the stakeholders.
- Based on the feedback and learning from various projects that have implemented the rating programme, the latest version 2.0 has been launched in March 2012. The rating system is designed to suit Indian climate and construction practices.

About 220 members representing 120 organisations have participated in the development of the rating programme. As on date (March 2012), over 400 million sq.ft. of built-up area is registered under the rating programme.

The rating system will be subject to review by the core committee, every year, to ensure that it is updated and contemporary.

- IGBC Green Homes Rating System is a voluntary and consensus-based programme.
- The rating system has been developed based on materials and technologies that are presently available.
- The objective of IGBC Green Homes Rating is to facilitate the creation of water efficiency, handling of house-hold waste, energy efficiency, healthy, comfortable and environmentally-friendly houses.
- IGBC Green Homes Rating System is a measurement system designed for rating new and major, renovated residential buildings which are broadly classified into two construction types:
  1. Individual residential unit
  2. Multi-dwelling residential units
     - ✓ Gated communities.
     - ✓ High-rise residential apartments.

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- ✓ Hostels, Service apartments, Resorts, Motels and Guest houses.

The rating programme will also undergo periodic revisions to incorporate the latest advances and changes.

IGBC Green Homes® Rating System

IGBC Green Homes Rating system addresses green features under the following categories:

1. Site Selection and Planning
2. Water Efficiency
3. Energy Efficiency
4. Materials & Resources
5. Indoor Environmental Quality
6. Innovation & Design Process

The guidelines detailed under each mandatory requirement & credit enables the design and construction of green homes of all sizes and types. Different levels of green building certification are awarded based on the total credits earned. However, every green home should meet certain mandatory requirements, which are non-negotiable.

The various levels of rating awarded are:

Certification Level	Recognition
Certified	Best Practices
Silver	Outstanding Performance
Gold	National Excellence
Platinum	Global Leadership

2.1.3.2 GRIHA - Green Rating for Integrated Habitat Assessment

- Human Habitats (buildings) interact with the environment in various ways. Throughout their life cycles, from construction to operation and then demolition, they consume resources in the form of energy, water, materials, etc. and emit wastes either directly in the form of municipal wastes or indirectly as emissions from electricity generation.
- GRIHA attempts to minimize a building's resource consumption, waste generation, and overall ecological impact to within certain nationally acceptable limits / benchmarks.
- GRIHA is a rating tool that helps people assesses the performance of their building against certain nationally acceptable benchmarks.

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The basic Features of GRIHA

A building is assessed based on its predicted performance over its entire life cycle – inception through operation. The stages of the life cycle that have been identified for evaluation are:

1. Pre-construction stage: intra- and inter-site issues like proximity to public transport, type of soil, kind of land, where the property is located, the flora and fauna on the land before construction activity starts, the natural landscape and land features.

2. Building planning and construction stages: issues of resource conservation and reduction in resource demand, resource utilization efficiency, resource recovery and reuse, and provisions for occupant health and well-being. The prime resources that are considered in this section are land, water, energy, air, and green cover.

3. Building operation and maintenance stage: issues of operation and maintenance of building systems and processes, monitoring and recording of energy consumption, and occupant health and well-being, and also issues that affect the global and local environment.

GRIHA – Rating Variants

1. ***SVAGRIHA Pre-Certification***

    - SVAGRIHA Precertification is applicable only for projects which are less than 2500 sq m built-up area. Any building, except for a factory building, is accepted under the SVAGRIHA Precertification rating system

2. ***SVAGRIHA***

    - SVAGRIHA can help in design and evaluation of individual residences, small offices, schools, motels, commercial buildings etc

3. ***GRIHA Rating***

    - All new construction projects with built up area more than 2500 m^2 (excluding parking, basement area, and typical buildings) are eligible for pre-certification

4. ***GRIHA AH Precertification***

    - All projects that are in-line with the PMAY requirements will be eligible for GRIHA AH since the rating strives to integrate sustainability in the residences in line with the PMAY.
    - All the upcoming projects that have approval/sanction letter issued by government agency (Central/State) confirming that the project is being developed as per Pradhan Mantri Awas Yojana (PMAY) are eligible for GRIHA AH rating.

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5. ***GRIHA for Existing Building Rating***
    - All operational buildings having a built-up area greater than 2,500 sq. m. are eligible for certification under GRIHA for Existing Buildings rating. The typology of buildings include: offices, retail spaces, institutional buildings, hotels, hospital buildings, healthcare facilities, residences, and multi-family high-rise buildings.
6. ***GRIHA for Existing Day Schools***
    - All day schools (excluding hostel/ residential facilities) which have been operational for at least 1 year are eligible for certification under GRIHA for Existing Day Schools rating.

7. ***GRIHA for Cities***

    - The GRIHA for CITIES rating has been structured as a framework for sustainable development of a city, to be achieved by measuring 'greenness' of existing as well as proposed cities .
    - The rating targets all cities, towns with civic/local bodies, such as Municipal Corporation and Municipal Council.

8. ***GRIHA for Large Developments***

    - The intent of GRIHA LD is to provide a consolidated framework for assessment of environmental impacts of large scale developments.

GRIHA – Benefits

This benefits the community at large with the improvement in the environment by reducing GHG (greenhouse gas) emissions, reducing energy consumption and the stress on natural resources.

Some of the benefits of a green design to a building owner, user, and the society as a whole are as follows:

1. Reduced energy consumption without sacrificing the comfort levels
2. Reduced destruction of natural areas, habitats, and biodiversity, and reduced soil loss from erosion etc.
3. Reduced air and water pollution (with direct health benefits)
4. Reduced water consumption
5. Limited waste generation due to recycling and reuse
6. Reduced pollution loads
7. Increased user productivity
8. Enhanced image and marketability

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2.2 Role of Green Building Rating Systems in Market Transformation:

The fundamental market transformation can be seen across key areas:

1. The Cost Premium & Feasibility

- Before Green Rating Systems: Building green was an expensive, experimental luxury. Because sustainable technologies were rare and unstandardized, developers faced a 10% to 15% upfront capital cost premium. Most eco-friendly technologies, specialized glass, or high-efficiency HVAC units had to be imported, making sustainability economically unviable for mainstream developers.
- After Green Rating Systems: The upfront cost premium has collapsed to a modest 1% to 3.6% for standard projects, with many developers achieving cost parity. The standardized rating points created massive demand, which drove local manufacturing economies of scale. These upfront costs are now easily recovered within 2 to 3 years through operational utility savings.

2. The Materials & Construction Supply Chain

- Before Green Rating Systems: The supply chain was highly fragmented. If a builder wanted "eco-friendly" options, they relied on unverified vendor claims. Basic items like traditional clay bricks (which destroy topsoil) and standard OPC cement were the absolute default.
- After Green Rating Systems: India now features a multibillion-dollar localized green building materials market. Rating systems forced structural accountability. Industrial waste byproducts like fly ash and slag are now standard components in concrete blends. Products like Autoclaved Aerated Concrete (AAC) blocks, low-VOC paints, and high-performance glass are locally manufactured commodities. Programs like CII's *GreenPro* provide formal labels so builders can instantly source certified eco-friendly materials.

3. Commercial Real Estate Demand & Tenant Behavior

- Before Green Rating Systems: Location, square footage, and exterior aesthetics were the only metrics that dictated "Grade A" office spaces. Energy and water efficiency were completely ignored during leasing negotiations because tenants viewed utility bills as an inevitable cost of doing business.
- After Green Rating Systems: Sustainability is a core corporate real estate requirement. With multinational corporations and Global Capability Centers (GCCs) bound by international ESG (Environmental, Social, and Governance) targets, the vast majority of India's Grade A office stocks are green-certified. Non-certified commercial properties now face a "brown discount"—meaning they suffer from higher vacancy rates and lower rental yields because modern corporate tenants actively filter them out.

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4. Architectural Design & Operational Data

- Before Green Rating Systems: "Green washing" ran rampant. A developer could plant a few trees or put glass on a building and market it as eco-friendly. Buildings were treated as static structures; sub-metering was rare, and real-time energy or water tracking was practically non-existent.
- After Green Rating Systems: Sustainability is now completely data-driven. To achieve certification, projects must verify true structural performance through smart sub-metering, Building Management Systems (BMS), and continuous Indoor Air Quality (IAQ) monitoring. The market shifted from subjective marketing claims to verified, empirical metrics: a certified building guarantees 20% to 30% lower energy use and 30% to 50% lower water consumption.

5. Public Policy & Government Regulations

- Before Green Rating Systems: Government building regulations (municipal bye-laws) were basic and focused entirely on setbacks, safety, and height restrictions. Environmental clearances were tedious, slow, and offered no rewards for sustainable design.
- After Green Rating Systems: Rating frameworks are deeply woven into public policy. They heavily influenced national guidelines like the Energy Conservation and Sustainable Building Code (ECSBC). Furthermore, state governments and municipal corporations actively incentivize these ratings. Today, achieving a high IGBC or GRIHA rating serves as a regulatory fast-track, granting developers concrete financial perks like 1% to 5% additional Floor Area Ratio (FAR) free of cost, property tax rebates, or streamlined environmental clearances.

2.3 Green Building Opportunities

Green buildings offer numerous opportunities for improving environmental sustainability, economic efficiency, and occupant well-being. They encourage the adoption of innovative technologies and resource-efficient practices throughout the building life cycle.

1. The Developer & Investor (The Business Case)

For those who finance and own real estate, green buildings have evolved from a "corporate social responsibility" box to check into a powerful tool for maximizing profits and protecting wealth.

- *Unlocking Global Capital:* International investors and major funds now operate under strict ESG (Environmental, Social, and Governance) rules. Certified green properties function like an elite passport, granting access to foreign direct investment (FDI) and low-interest "green loans" that aren't available to standard projects.
- *Higher Yields, Lower Risk:* High-profile corporate tenants are willing to pay a premium for certified Gold or Platinum spaces to meet their own carbon-reduction

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targets. These buildings fill up faster, command higher rent, and experience much lower tenant turnover.

- *Avoiding Asset Obsolescence:* As environmental regulations tighten, traditional buildings face a "brown discount"—they become more expensive to run and harder to lease. Going green future-proofs the property's long-term market value.

2. The Architect, Civil Engineer, & Consultant (The Technical Edge)

For design and construction professionals, the green building shift moves engineering away from standard, repetitive practices and pushes it toward high-value, data-driven innovation.

- *The Rise of Smart Tech:* Earning top rating points requires embedding cutting-edge technology directly into the building's skeleton. This opens massive opportunities in:
    - *BIM (Building Information Modelling) & Digital Twins:* Using advanced software to simulate daylight, airflow, and energy use before a single brick is laid.
    - *AI & IoT Integration:* Designing automated systems where smart sensors adjust cooling, lighting, and ventilation based on real-time room occupancy and Carbon di oxide levels.
- *High-Demand Career Niches:* Navigating complex rating systems has created a lucrative job market for specialized professionals. There is a soaring demand for Certified Energy Auditors, Life Cycle Assessment (LCA) experts, and Accredited Professionals (like IGBC AP or GRIHA CP) who manage the certification process.

3. The Manufacturer & Materials Vendor (The Industrial Boom)

For companies that produce and supply building products, rating systems have completely rewritten the rules of what materials a job site requires.

- *Massive Green Supply Chain Market:* Rating credits heavily reward projects that buy local, recycled, and eco-certified materials. This has opened up a multibillion-dollar domestic market for manufacturers producing items like:
    - High-performance low-E glass and advanced insulation.
    - Eco-friendly walling like Autoclaved Aerated Concrete (AAC) blocks and fly-ash bricks.
    - Low-VOC paints, sustainable adhesives, and recycled-content flooring.
- *Water & Waste Tech Innovation:* Because rating systems demand drastic reductions in water usage, there is an absolute boom for companies selling localized Sewage Treatment Plants (STPs), smart greywater recycling systems, and ultra-efficient plumbing hardware.

4. Policy Makers & Municipalities (The Urban Strategy)

For local and state governments, green building frameworks provide a pre-made toolkit to upgrade city infrastructure without breaking the public budget.

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- *Leveraging Public Incentives:* Governments can drive massive environmental changes simply by offering structural perks instead of direct cash hand-outs. Today, developers targeting high ratings can secure:
    - Extra Floor Area Ratio (FAR): Getting 1% to 5% extra buildable space for free, this instantly increases their saleable square footage.
    - Fast-Track Approvals: Bypassing slow municipal bottlenecks and moving through environmental clearances at high speed.
- Relieving Strain on City Infrastructure: By rewarding on-site solar energy production, rainwater harvesting, and waste recycling, green buildings act as self-sustaining pockets. This drastically reduces the daily load on city power grids, municipal water lines, and crowded landfills.

2.4 Green Building Features

2.4.1 Material and Resources

1. *Separation of House-hold Waste*
    - *IGBC Intent:* Facilitate segregation of house-hold waste at source so as to prevent such waste being sent to land-fills.
    - *Approach and Methodologies:*
      Allocate suitable space on-site for sorting out dry and wet waste. Examine the scope for recycling of waste collected from the project. Identify local dealers to collect and dispose waste material such as paper, plastic, metals, glass, cardboard, organic waste, batteries, E-waste and lamps. Educate residents about various recycling methods.
2. *Organic Waste Management, Post-occupancy*
    - *IGBC Intent:* Ensure effective organic waste management, post-occupancy, so as to prevent waste being sent to land-fills.
    - *Approach and Methodologies:* Calculate approximate quantity of organic waste generated in the building. Design the organic waste treatment plant with suitable capacity to efficiently manage the waste. Typical technologies recognised include digesters, gasifiers, vermi-composting, etc.,
3. *Handling of Construction Waste Materials*
    - *IGBC Intent:* Encourage practices to manage construction waste, thereby, avoiding waste being sent to land-fills.
    - *Approach and Methodologies:* Collect all construction debris generated on-site. Segregate this waste based on their utility. Examine means of reusing such waste within the project or other projects, (or) identify appropriate vendors to divert such waste to manufacturing units which would use them as raw materials. Typical construction debris in residential projects include broken bricks, steel bars, broken tiles, glass, wood waste, paint cans, cement bags, packing materials, etc.
4. *Reuse of Salvaged Materials*

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- *IGBC Intent:* Encourage the use of salvaged building materials and products to reduce the demand for virgin materials, thereby, minimising the impacts associated with extraction and processing of virgin materials.
- *Approach and Methodologies:* Identify opportunities to incorporate salvaged materials in the building(s) / campus. Consider using salvaged materials for applications such as bricks, doors, flooring, frames, furniture, panelling, etc.,

5. *Materials with Recycled Content*
   - *IGBC Intent:* Encourage use of materials which contain recycled content to reduce environmental impacts associated with the use of virgin materials.
   - *Approach and Methodologies:* Survey the materials with recycled content identify such local suppliers. Materials with recycled content include aluminium, cement, composite wood, concrete, glass, gypsum, false ceiling, fly ash blocks, steel, tiles, etc.,
6. *Local Materials*
   - *IGBC Intent:* Encourage use of building materials available locally, thereby, minimising the associated environmental impacts resulting from transportation.
   - *Approach and Methodologies:* Survey and identify building materials which are in the specified radius, in early stages of project design. While selecting local materials, ensure that they perform better in terms of strength, maintenance and durability.
7. *Rapidly Renewable Building Materials & Certified Wood*
   - *IGBC Intent:* Minimise use of virgin wood, thereby, encouraging responsible forest management and maximise use of materials which are rapidly renewable.
   - *Approach and Methodologies:* Identify all wood-based applications in the project. Determine the types of materials/ products needed (e.g., doors, windows, furniture, flooring etc.,). Explore the possibility of using rapidly renewable materials and wood certified by Forest Stewardship Council (FSC) or Programme for the Endorsement for Forest Certification (PEFC) or equivalent, for all such wood-based applications. Also, while sourcing wood for various applications, specify the quality or grade of wood required. Locate local suppliers so as to reduce additional costs and environmental impacts caused during transportation.

2.4.2 Water Efficiency

1. *Rainwater Harvesting, Roof & Non-roof*
   - *IGBC Intent:* Enhance ground water table and reduce municipal water demand through effective rain water management.
   - *IGBC Approach and Methodologies:* Survey the water table in the project's location. Design appropriate harvesting system based on the sub surface characteristics. Factors to be considered include weathering, fractures & joints for rocky sites and thickness of aquifer for sedimentary sites.

     Capture rainwater from roof top for reuse. The design should also include flushing arrangement to let out impurities in the first few showers. Such pollutants and impurities include paper waste, leaves, bird droppings, dust, etc.,

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2. *Water Efficient Plumbing Fixtures*
    - *IGBC Intent:* Minimise the use of municipal water and reduce load on waste water systems.
    - *Approach and Methodologies:* While selecting water fixtures, look for their efficiencies. The product catalogue or the brochure may detail the flow rates at various pressures. Fixtures are available with ultra-high efficiency which can reduce substantial quantity of water consumption.

      Consider reuse of treated waste water and captured rain water for flushing. Ensure periodic testing of the treated water to meet the quality standards for flushing, as prescribed by Central / State Pollution Control Board.
3. *Landscape Design*
    - *IGBC Intent:* Design landscape to ensure minimum water consumption.
    - *Approach and Methodologies:* During landscape design minimise turf to the extent possible. Select ground covers, shrubs and trees which consume less water. Select species that are well-adapted to the site and which are drought tolerant. Consider xeriscaping as an approach for landscaping.
4. *Management of Irrigation Systems*
    - *IGBC Intent:* Reduce water demand for irrigation through water efficient management systems and techniques.
    - *Approach and Methodologies:* Design irrigation management system based on the requirements of the landscape plan. Conduct a market survey on the technologies available to manage irrigation efficiently. The designer and the installer must work together and ensure the planned performance of the system. Also, identify local manufacturers supplying systems such as moisture sensors, time-based controllers, etc.
5. *Waste Water Treatment and Reuse*
    - *IGBC Intent:* Reduce consumption of potable water and waste water generation to minimise the burden on municipal streams.
    - Approach and Methodologies: Calculate the waste water volumes generated in the building. Design appropriately the capacity of the on-site waste water treatment system. While designing the treatment system, ensure that the treated waste water meets the required quality standards based on its purpose of application. Ensure that the quality of the treated waste water is fit and safe for reuse.

      Prioritise the use of treated waste water such that irrigation, flushing, etc., Excess treated waste water can also be used for make-up water for air-conditioning systems and other purposes. Ensure periodic testing of the treated water to meet the quality standards as prescribed by Central / State Pollution Control Board. Have adequate signages all around the building to caution residents and housekeeping staff that this water is not potable.
6. *Water Metering*
    - *IGBC Intent:* Encourage continuous monitoring to enhance the water performance in the project.

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- *Approach and Methodologies:* Identify all the major water consuming areas and install systems to monitor their consumptions. Develop and implement a measurement and verification mechanism to compare predicted water savings to actual water consumption.

2.4.3 Optimum Energy Efficiency

1. *CFC-free Equipment*
   - *IGBC Intent:* Avoid use of refrigerants and ozone depleting gases which has negative impact to the environment.
   - *Approach and Methodologies:* Survey the market for CFC-free HVAC system/ unitary air-conditioners. Install HVAC equipment/ unitary air-conditioners which does not use CFC based refrigerant.
2. *Minimum Energy Performance*
   - *IGBC Intent:* Improve energy efficiency of the building(s) to reduce environmental impacts from excessive energy use.
   - *Approach and Methodologies:* The approach would be to look at orientation, right at the design stage. The design should then consider and select appropriate measures such as insulation of walls, roof, selection of glass, sizing of windows, lighting and efficient air-conditioning & heating systems.

     Identify the materials and equipment available in the market and their properties with regard to energy performance. While selecting these materials and equipment, consider their associated environmental impacts.

     Also, consider the renewable energy feasibility while determining the power requirement from the local utility. The renewable energy sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,
3. *Enhanced Energy Performance*
   - *IGBC Intent:* Enhance energy efficiency of the building(s) to reduce environmental impacts from excessive energy use.
   - *Approach and Methodologies:* The approach would be to look at orientation, right at the design stage. The design should then consider and select appropriate measures such as insulation of walls, roof, selection of glass, sizing of windows, lighting and efficient air-conditioning & heating systems.

     Identify the materials and equipment available in the market and their properties with regard to energy performance. While selecting these materials and equipment, consider their associated environmental impacts.

     Also, consider the renewable energy feasibility while determining the power requirement from the local utility. The renewable energy sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,
4. *On-site Renewable Energy*
   - *IGBC Intent:* Promote self-sufficiency in energy through renewable technologies for on-site power generation and use within the project.
   - *Approach and Methodologies:* Consider renewable energy feasibility while determining the power requirement from the local utility. Renewable energy

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sources that can be considered are - Solar energy, Wind energy, Biomass, Bio-gas etc.,

5. *Solar Water Heating System*
   - *IGBC Intent:* Encourage use of solar energy for water heating applications in the building(s).
   - Approach and Methodologies: Calculate hot water requirement for the entire project. Install solar water heating system, thereby catering to hot water requirement in the project. The design should also consider the availability of space to install the solar panels.
6. *Energy Saving Measures in Appliances & Other Equipment*
   - *IGBC Intent:* Conserve energy in the use of house-hold appliances and other equipment, thereby reducing environmental impacts.
   - *Approach and methodology:* Conduct market survey before selecting appliances / equipment. Consider the energy effi ciency of this appliances / equipment and select minimum BEE 4-star rated or equivalent appliances / equipment.
7. *Distributed Power Generation*
   - *IGBC Intent*: Reduce dependence on fossil fuels for in-situ power generation.
   - *Approach and Methodologies:* Survey the market for available distributed power generators which operate on bio-fuels or non-edible oils, and diesel generator sets which are CPCB compliant for emissions & noise. Also, consider availability of fuel on a sustained basis and economic viability of the system.
8. *Energy Metering*
   - *IGBC Intent:* Encourage continuous monitoring to enhance the energy performance in the project
   - *Approach & Methodologies:* Identify all the major energy consuming equipment and install systems to monitor energy consumption. Develop and implement a measurement & verification mechanism to compare predicted savings to actual energy performance of the installed meters.

2.5 Typical Energy Saving Approach in Buildings

Energy-saving approaches in buildings focus on reducing energy consumption while maintaining thermal comfort, indoor air quality, and occupant productivity. These strategies can be incorporated during the design, construction, and operational stages of a building.

1. Building Orientation

Proper orientation minimizes heat gain in summer and maximizes natural daylight and winter solar heat. Orienting the longer side of the building along the east-west direction reduces direct solar radiation and lowers cooling loads.

2. High-Performance Building Envelope

A well-insulated roof, walls, and floors, along with energy-efficient windows and doors, reduce unwanted heat transfer. Double-glazed windows, insulated walls, and airtight

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construction help maintain indoor temperatures with less reliance on mechanical heating and cooling.

3. Passive Solar Design

Passive solar techniques utilize natural sunlight for heating and lighting. Features such as shading devices, overhangs, thermal mass, and strategically placed windows reduce artificial energy requirements throughout the year.

4. Natural Ventilation

Cross ventilation, stack ventilation, and well-designed openings improve indoor air circulation and reduce dependence on air conditioning. Proper airflow also enhances indoor comfort and air quality.

5. Day lighting

Maximizing the use of natural daylight through skylights, clerestory windows, light shelves, and large glazed openings reduces the need for artificial lighting during daytime while improving occupant well-being.

6. Energy-Efficient Lighting

Replacing conventional lamps with LED lighting and installing occupancy sensors, daylight sensors, and automatic lighting controls significantly lowers electricity consumption.

7. Efficient HVAC Systems

High-efficiency heating, ventilation, and air-conditioning (HVAC) systems consume less energy while maintaining thermal comfort. Proper sizing, regular maintenance, and programmable thermostats further improve system efficiency.

8. Renewable Energy Integration

Installing rooftop solar photovoltaic (PV) systems, solar water heaters, and other renewable energy technologies reduces dependence on conventional electricity and lowers greenhouse gas emissions.

9. Energy-Efficient Appliances and Equipment

Using appliances with high energy ratings, variable-speed motors, and efficient pumps and fans minimizes electricity consumption without compromising performance.

10. Building Energy Management Systems (BEMS)

Smart energy management systems monitor and control lighting, HVAC, and other electrical systems in real time. Automated controls optimize energy use based on occupancy and environmental conditions.

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11. Water Heating Efficiency

Energy consumption for water heating can be reduced by using solar water heaters, heat pump water heaters, insulated hot-water pipes, and low-flow fixtures.

12. Landscaping and Green Roofs

Trees, vegetation, green roofs, and vertical gardens provide shading , reduce the urban heat island effect, and lower surrounding air temperatures, thereby decreasing cooling energy demand.

13. Regular Maintenance and Energy Audits

Periodic maintenance of building systems and energy audits help identify inefficiencies, equipment faults, and opportunities for energy conservation, ensuring long-term building performance.

14. Occupant Awareness and Energy-Conscious Practices

Educating occupants about energy-saving habits, such as switching off unused lights and equipment, maintaining appropriate thermostat settings, and using daylight effectively, contributes significantly to overall energy savings.

2.6 LEED India Rating System and Energy Efficiency

2.6.1 LEED– Leadership in Energy and Environmental Design

- *Leadership in Energy and Environmental Design (LEED)* is a green building certification program used worldwide.
- Developed by the non-profit U.S. Green Building Council (USGBC), it includes a set of rating systems for the design, construction, operation, and maintenance of green buildings, homes, and neighborhoods, which aims to help building owners and operators be environmentally responsible and use resources efficiently.
- According to LEED itself, “The LEED plaque on a building is a mark of quality and achievement in green building.”
- As such, any building with LEED certification has proven itself to be designed, built, and maintained with the highest degree of energy efficiency.
- There are a number of benefits of LEED certification such as:
    1. Lower operating costs;
    2. Reduction in risks and liability;
    3. More durable buildings;
    4. Better indoor air quality;
    5. Improved resale value; and
    6. Support for the local economy.
- LEED projects are typically scored across nine areas that each speaks to a specific aspect of green building design:

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1. Integrative process;
2. Location and transportation;
3. Sustainable sites;
4. Water efficiency;
5. Energy and atmosphere;
6. Materials and resources;
7. Indoor environmental quality;
8. Innovation; and
9. Regional Priority.

- This rating system applies to new construction projects, existing buildings, neighborhood developments, and even cities and communities.
- Each certification will be valid for three years, at the end of which the project will need to get recertified again.
- If your project qualifies for a LEED certificate, it will earn one of the following four ratings:

    i. Platinum: 80-plus points earned;
    ii. Gold: 60-79 points earned;
    iii. Silver: 50-59 points earned; or
    iv. Certified: 40-49 points earned.

- This rating system applies to new construction projects, existing buildings, neighborhood developments, and even cities and communities.

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UNIT -3

GREEN BUILDING DESIGN

3.1 Introduction to Green Building Design

Green building design, also known as **sustainable building design**, is an approach to constructing and operating buildings in a way that reduces negative environmental impacts and enhances human health, comfort, and efficiency throughout the building's life cycle. It emphasizes the responsible use of resources such as energy, water, materials, and land, while minimizing waste and pollution.

A **green building** is designed to:

- **Use less energy and water** than conventional buildings.
- **Reduce waste** during construction and operation.
- **Utilize eco-friendly and recycled materials.**
- **Improve indoor environmental quality** for occupants.
- **Integrate renewable energy sources** such as solar or wind.
- **Protect biodiversity** and reduce strain on natural ecosystems.

Key Principles of Green Building Design

1. **Sustainable Site Planning** – choosing locations that minimize environmental disruption.
2. **Efficient Energy Use** – reducing consumption through insulation, daylighting, natural ventilation, and renewable energy systems.
3. **Water Efficiency** – low-flow fixtures, rainwater harvesting, and wastewater recycling.
4. **Materials and Resources** – use of recycled, rapidly renewable, and low-impact building materials.
5. **Indoor Environmental Quality** – ensuring good air quality, natural lighting, and thermal comfort.
6. **Waste Reduction** – adopting construction practices that minimize waste and encourage reuse/recycling.
7. **Life-Cycle Approach** – considering the environmental impact of materials and systems over the building's entire lifespan.

Benefits of Green Building Design

- **Environmental:** Reduced greenhouse gas emissions, conservation of natural resources.
- **Economic:** Lower operating costs due to energy and water savings.

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- **Health & Social:** Better indoor air quality, improved comfort and productivity of occupants.

3.2 Reduction in energy demand:

1. **Building Orientation & Design** – Proper orientation, shading devices, and passive solar design reduce heating and cooling loads.
2. **Efficient Insulation** – High-quality wall, roof, and floor insulation minimizes heat loss or gain, lowering energy needs.
3. **Natural Ventilation** – Designing for cross-ventilation reduces dependence on mechanical cooling systems.
4. **Daylighting** – Maximizing natural light through windows, skylights, and reflective surfaces cuts down on artificial lighting demand.
5. **Energy-Efficient Appliances** – Using star-rated appliances and LED lighting lowers overall energy consumption.
6. **Smart Controls & Automation** – Sensors, timers, and smart thermostats optimize energy use based on occupancy.
7. **Renewable Energy Integration** – Solar panels, wind turbines, and other renewable systems reduce reliance on grid electricity.
8. **Efficient HVAC Systems** – Installing high-performance heating, ventilation, and cooling systems reduces operational energy demand.
9. **Water-Energy Nexus** – Water-saving technologies (low-flow fixtures, rainwater harvesting) indirectly reduce energy used for pumping and treatment.
10. **Behavioral Changes** – Encouraging energy-conscious habits, such as switching off unused devices, greatly reduces overall demand.

3.3 Onsite sources and sinks

Onsite Sources (Inputs/Resources available within the site)

1. **Solar Energy** – Rooftop solar panels capture sunlight to generate electricity or heat water.
2. **Rainwater** – Collected through harvesting systems for non-potable uses like flushing and irrigation.
3. **Wind Energy** – Small wind turbines harness natural airflow for power generation.
4. **Geothermal Energy** – Heat from the ground can be used for space heating and cooling.
5. **Biomass & Organic Waste** – Kitchen and garden waste can be converted into biogas or compost.

Onsite Sinks (Outputs/Ways to absorb or manage waste and emissions onsite)

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6. **Green Roofs & Landscaping** – Act as carbon sinks, absorb CO_2, and reduce heat island effect.
7. **Rain Gardens & Pervious Surfaces** – Allow stormwater absorption, reducing runoff and groundwater depletion.
8. **Greywater Recycling Systems** – Treat wastewater from sinks/showers for reuse onsite.
9. **Waste-to-Energy Units** – Convert solid waste into energy, reducing landfill burden.
10. **Carbon Sequestration through Vegetation** – Trees and plants absorb CO_2, improving local air quality.

3.4 maximizing system efficiency

Maximizing system efficiency in green buildings involves an integrated approach combining advanced technologies, optimized design, smart controls, and proactive management to reduce resource use and boost performance.

Key Strategies to Maximize Efficiency

High-Performance Building Design

- Use passive design (proper orientation, natural ventilation, daylighting) to cut energy use for lighting and temperature control.
- Insulate and seal the building envelope with high-performance windows and materials to minimize thermal losses.

Advanced HVAC and Mechanical Systems

- Install energy-efficient HVAC systems like variable refrigerant flow (VRF), geothermal heat pumps, and smart thermostats, ensuring optimal temperature with minimal energy.
- Regularly maintain and commission mechanical systems to sustain peak efficiency throughout the building's life.

Smart Energy Management and Controls

- Implement Building Energy Management Systems (BEMS) that monitor, analyze, and adjust energy usage in real time.
- Utilize IoT sensors for occupancy, light, and environmental detection, enabling dynamic control over HVAC, lighting, and appliances.
- Automate controls for lighting, blinds, and HVAC based on occupancy and natural environmental conditions.

Lighting and Appliances Optimization

- Replace all outdated fixtures with LED lighting and install daylight and occupancy sensors to eliminate unnecessary energy use.

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- Use energy-efficient appliances and devices with high efficiency ratings throughout the building.

Water and Waste Management

- Incorporate low-flow fixtures, rainwater harvesting, and greywater recycling systems to minimize water consumption and lower the building's overall energy demand.

Renewable Energy Integration

- Install solar panels, wind turbines, or geothermal systems to reduce dependency on nonrenewable energy and support net-zero targets.
- Couple renewables with smart energy management for optimal use and storage.

Continuous Monitoring and Improvement

- Collect and analyze energy performance data to identify inefficiencies and guide future upgrades, maintenance, and occupant education.
- Apply predictive analytics and cloud-based systems for remote monitoring and fast response to performance issues.

Facility Management's Role

- Expert facility management optimizes daily operations and ensures all systems run smoothly, maximizing efficiency and sustainability outcomes.

Implementing these strategies leads to significantly lower energy bills, reduced environmental impact, and enhanced occupant comfort, making green buildings efficient, resilient, and future-ready.

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3.5 Steps to reduce energy demand and use onsite sources and sinks

Detailed Steps to Reduce Energy Demand

1. Energy Audits and Monitoring

- Conduct an **energy audit** using smart meters or energy management systems to identify where most energy is being consumed.
- Real-time monitoring allows prioritization of key areas for improvement and helps set measurable reduction targets.

2. Upgrade to Efficient Equipment

- Replace old appliances and machinery with **energy-efficient models** (e.g., Energy Star-rated, or A+++).
- Upgrade all lighting to **efficient LEDs**, and use automated lighting controls (sensors, timers).

3. Schedule and Automate Loads

- **Stagger the operation** of high-energy-use equipment to prevent simultaneous consumption spikes (reduce peak demand).
- Use scheduled controls or building automation systems so that not all HVAC, lighting, or machines run at once.
- Introduce **smart automation** (IoT devices) that adjusts operation based on occupancy or time of day.

4. Optimize HVAC and Building Envelope

- Insulate walls, roofs, and seal windows/doors to minimize heating/cooling losses.
- Install **programmable thermostats** and zoning; regularly service HVAC equipment for efficient operation.
- Use **passive design** where possible—shading, natural ventilation, and daylighting.

5. Behavioral Change and Education

- Educate staff and building occupants about energy-saving habits.
- Encourage shutting off unused equipment and reporting inefficiencies.

6. Load Shedding and Shifting

- Temporarily turn off noncritical loads during peak periods (load shedding).
- Shift energy-intensive tasks—such as manufacturing or charging— to off-peak periods (load shifting).

7. Retrofit and Improve Building Fabric

- Retrofitting old buildings with better insulation, glazing, and air-tightness standards is essential for long-term reduction.ukerc.rl

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Using Onsite Energy Sources and Sinks

1. Onsite Renewable Energy Generation

- Install **solar PV panels** to generate electricity; size systems to meet a proportion of demand onsite.
- Use **solar water heaters** or air heaters for hot water and space conditioning needs.

2. Energy Storage

- Use **batteries** (electrical storage) to store power generated during low-demand periods, supplying it during peak times (peak shaving).
- Consider thermal storage, such as **ice storage** in HVAC systems, to shift cooling load to off-peak hours.

3. Heat Recovery and Reuse

- Install **waste heat recovery** systems in industrial or commercial facilities, capturing heat from processes or exhausts for use elsewhere.

4. Green Infrastructure and Passive Sinks

- Use **green roofs**, shade trees, and reflective materials to reduce heat gain and lower cooling loads.
- Employ landscaping and living walls for natural cooling effects and rainwater management.

5. Integrate Demand Response and Smart Controls

- Join **demand response programs** to reduce demand when the grid is under stress, often earning incentives for doing so.
- Use **smart controls** (IoT solutions) to shift or reduce loads automatically based on real-time factors like occupancy, temperature, or grid signals.

Underlying Principles: Avoid, Shift, Improve

- **Avoid** unnecessary energy use (e.g., extend product lifespans, turn off unused devices).
- **Shift** to less energy-intensive practices/times (e.g., use district heating, schedule energy use to off-peak hours).
- **Improve** efficiency of equipment and buildings (retrofitting, high-efficiency systems).

A holistic approach—combining technical upgrades, behavioral change, smart automation, renewables, and building improvements—produces the greatest, most sustainable reduction in energy demand and maximizes the use of available onsite energy resources and sinks.

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3.6 Use of Renewable energy sources

The use of **renewable energy sources in green buildings** is essential for achieving sustainability, low-carbon operation, and improved energy resilience. Green buildings integrate various renewable technologies to generate clean energy, reduce environmental impact, and meet demanding efficiency standards.

Types of Renewable Energy Used in Green Buildings

Solar Energy

- **Photovoltaic (PV) Panels:** These are widely integrated into rooftops, facades, or even window glass. They convert sunlight directly into electricity, and advances such as monocrystalline, polycrystalline, and thin-film panels provide varied efficiencies and design options. PV technology has become more affordable and efficient with technological progress.
- **Solar Thermal Systems:** These systems capture solar energy for heating water or air, supporting hot water supply, space heating, and sometimes cooling. Solar collectors come in flat plate, evacuated tube, or integrated systems suitable for various applications.

Wind Energy

- **Small Wind Turbines:** These can supplement the energy needs of buildings in areas with sufficient wind resources. Innovations include compact vertical-axis turbines, which are suitable for urban environments due to their ability to capture wind from any direction and operate in limited space.
- Integration of wind systems in green building design complements solar generation and can significantly boost energy self-sufficiency, especially in hybrid systems.

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Geothermal Energy

- **Geothermal Heat Pumps (GHP):** These systems exploit the earth's steady subsurface temperature to provide high-efficiency heating and cooling. They can use either horizontal or vertical configurations depending on land availability and site conditions, and deliver significant energy savings compared to conventional systems.
- **Direct Geothermal Use:** Where local geothermal resources are present, they can supply hot water or space heating directly.

Bioenergy

- **Biomass Boilers and Combined Heat and Power (CHP):** These use organic materials (wood pellets, agricultural waste, or biogas) to provide heat and sometimes electricity. Bioenergy systems are reliable, produce steady outputs, and can run on local organic waste, supporting both sustainability and waste management.

Hydropower (Micro-Scale)

- **Micro-hydropower:** Suitable for buildings close to water sources. These systems use water flow to generate electricity, further diversifying renewable options when site conditions allow.

Benefits and Opportunities

- **Carbon Reduction:** All renewable systems drastically reduce operational carbon emissions, contributing directly to sustainability and climate change goals.
- **Energy and Cost Savings:** Renewables lower long-term energy bills and protect against price volatility of fossil fuels. Financial incentives like tax rebates and subsidies further improve project viablity.
- **Self-Sufficiency and Resilience:** Generating energy onsite enhances independence from external supply disruptions and increases the resilience of the building.
- **Innovation and Flexibility:** Advances in solar, wind, and hybrid systems, along with smart building management, allow for tailored and integrated solutions even in dense urban sites.

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3.7 Eco friendly captive power generation for factory

Eco-friendly captive power generation for factories typically involves integrating renewable energy sources such as solar, wind, or biomass directly on-site to supply reliable, sustainable electricity and significantly reduce carbon emissions.

Leading Green Captive Power Options

- **Solar Captive Power Plants:** Factories increasingly install solar photovoltaic (PV) panels (either rooftop or ground-mounted) on-site to generate electricity for their own use, ensuring energy independence, lower costs, and reduced carbon footprint. Industries can also benefit from the group captive solar model, sharing generation and investment among several companies.

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- **Wind Power Systems:** Wind turbines can be installed within or near industrial campuses where wind conditions are suitable, offering a renewable source with zero emissions and operational cost benefits over time.
- **Biomass Power:** Utilizing agricultural or organic waste, factories can run biomass power plants, converting waste into energy for on-site use. This helps with waste management and supports the circular economy.
- **Combined Heat and Power (CHP) with Renewables:** Advanced captive systems now frequently employ co-generation (CHP), making use of waste heat from generation processes for heating or industrial tasks, resulting in efficiency as high as 80%.

Modern Enhancements and Hybrid Models

- **Energy Storage:** Integrated battery energy storage systems (such as lithium-ion batteries) stabilize the intermittency of solar and wind, ensuring a consistent supply during downtime and peak demand.
- **Hybrid Systems:** Many modern captive power plants combine multiple renewables—such as solar plus wind, or solar plus biomass—to balance supply and enhance reliability.
- **Emerging Technologies:** Innovations like perovskite solar cells and green hydrogen have potential for even greater sustainability and efficiency in future captive power solutions.

Advantages for Factories

- Reduction in energy costs and grid dependency.
- Enhanced energy security and operational resilience.
- Compliance with environmental regulations and improved sustainability credentials.
- Possibility of tax benefits and incentives, especially for solar and other renewables.

Considerations

While the initial investment in eco-friendly captive power generation may be higher, the long-term benefits in operational savings, reliability, and carbon emission reduction make it an increasingly favored option for manufacturing and heavy industry.

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3.8 Building requirements

Building requirements for an eco-friendly captive power plant in a factory include careful planning, infrastructure readiness, regulatory compliance, and integration of renewable technology to ensure reliable and cost-effective energy generation.

Key Infrastructure Needs

- **Site Preparation:** Land leveling, laying foundations, and civil works (for solar panels, wind turbines, or biomass units) are needed before installation.
- **Equipment Installation:** Space is required for core generation assets (solar panels, wind turbines, biomass digesters, or other), electrical wiring, mounting structures, inverters, safety controls, and monitoring systems.
- **Grid and Loads:** Grid interconnection infrastructure may be necessary, with sufficient room for transformers, substations, net metering, and energy storage (batteries, if included).

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- **Permits and Regulations:** Factories must secure all required statutory approvals—such as environmental clearance, building permits, and grid connectivity agreements—before construction begins.

Legal and Regulatory Requirements

- **Ownership and Consumption:** As per Indian law, at least 26% of the plant ownership must reside with the captive users, and at least 51% of generated power must be consumed by these users annually, to qualify as a captive plant.manupatra+2
- **Compliance Standards:** The plant must adhere to environmental norms, safety standards, and relevant state/central electricity regulations.manupatra+1
- **Open Access Rights:** Captive generators have rights to open access for transmitting self-generated power, subject to transmission facility availability.eqmagpro+1

Project Phases

- **Feasibility and Design:** Assess energy profile, choose the most suitable renewable technology, and design the appropriate capacity.
- **Engineering and Procurement:** Select certified suppliers and procure all necessary components.
- **Construction and Commissioning:** Execute civil works, install systems, conduct trials, and certify plant operation before regular use.

By following these building and regulatory requirements, factories can successfully implement eco-friendly captive power plants while ensuring compliance and long-term operational benefits.manupatra+2

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Unit 4

4.1 Introduction of air conditioning

Air conditioning is the process of controlling the temperature, humidity, purity, and movement of air in an enclosed space to create a comfortable indoor environment, regardless of external conditions.

History and Purpose

The modern air conditioner was invented in 1902 by Willis Carrier to improve air quality and control humidity for a printing company. Early systems evolved from simple methods, like hanging wet mats for evaporative cooling, to sophisticated refrigeration-based devices used globally today.

Basic Operation and Components

Air conditioners work by removing heat from indoor air and transferring it outside. The main components include:

- Compressor
- Condenser coil
- Evaporator coil
- Refrigerant fluid

A fan draws warm indoor air over the evaporator coil, where the refrigerant absorbs heat and turns from a liquid into a gas. The compressor then pressurizes this gas, which releases the absorbed heat to the outdoors via the condenser coil before the refrigerant cycles back, starting the process again.

Types of Air Conditioners

- Central air conditioning systems are common in large buildings, using ducts to distribute cooled air.
- Split-systems have interior and exterior units and are widely used in homes and offices.
- Packaged systems combine all components in a single outdoor unit.
- Portable and window units are used for single rooms or smaller spaces.

Additional Functions

Air conditioners also dehumidify the air, improve air quality through filtration, and can be configured for both cooling and heating in heat pump systems.

In summary, air conditioning has become essential for modern living, enabling comfort and productivity by precisely managing indoor climates in residential, commercial, and industrial settings.

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4.2 CII Godrej Green business centre

The CII Sohrabji Godrej Green Business Centre (CII-GBC) is a pioneering green building and business centre located in Hyderabad, Telangana, India. It was established in 2004 as a joint initiative between the Confederation of Indian Industry (CII) and the Godrej Group, with the aim to promote sustainable development and green business practices across India.

Key Features and Achievements

- The CII-GBC building is one of the world's greenest buildings and was the first in India to receive the prestigious LEED Platinum rating, and the third such building in the world at the time of certification in 2003.
- Its design incorporates sustainable architecture, including a circular footprint for best use of space and natural ventilation, courtyards, green walls, roofs, and a wind tower system to reduce energy use.
- Extensive use of recycled and eco-friendly building materials, on-site renewable energy systems, rainwater harvesting, and passive design practices make it a model for environmental sustainability.

Role and Activities

- The centre offers advisory services in green buildings, energy efficiency, environmental management, renewable energy, and climate change.
- It provides training, certification, and resources to businesses interested in adopting green practices and has played a key role in spreading the green building movement through the Indian Green Building Council (IGBC).
- The centre regularly hosts events, training programs, and national awards to promote energy efficiency and sustainability leadership in industry.

Contact and Address

- Address: Survey #64, Kothaguda Post, Ranga Reddy District, Hyderabad – 500084, India.
- The centre serves as a landmark both for its architecture and for its impact on sustainable development across India.

In summary, the CII Godrej Green Business Centre stands as a benchmark for green buildings in India, driving leadership and innovation in sustainable business and construction.

4.3 design philosophy

The design philosophy of the CII Sohrabji Godrej Green Business Centre in Hyderabad centers on sustainable architecture that gives back to nature what it takes. The building occupies just 9.2% of the 5-acre site to minimize environmental disruption and harmonize with the natural landscape. Its circular form follows the site's contours and rock formations to maximize natural ventilation and solar gain. A central courtyard connects functional spaces while enhancing natural light and cross-ventilation, fostering thermal comfort and occupant

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well-being. The design integrates traditional Indian climate control elements such as jaali screens for air and light diffusion and wind towers for passive cooling.

4.4 Design Interventions

- Green Roofs and Vegetation: Over 50% of the roof area is vegetated, reducing heat island effect and insulating the building.
- Solar Photovoltaic System: A 24 kW solar array on the roof supplies about 20% of the building's energy needs.
- Passive Cooling Techniques: Use of wind towers and open courtyards facilitate natural ventilation; evaporative air cooling reduces interior temperatures by up to 8°C.
- Water Conservation: Rainwater harvesting and water-efficient fixtures reduce water consumption by approximately 35%.
- Material Sustainability: Predominantly local and recycled building materials like fly-ash bricks, aerated concrete blocks, recycled glass, and aluminum reduce embodied energy. Interiors use organic materials such as bagasse-based composites.
- Daylighting and Thermal Comfort: Around 90% of interior spaces receive direct sunlight with double-glazed, argon-filled windows to optimize daylighting while minimizing heat gain.
- Integrated Building Management System: Automation controls lighting, ventilation, and power to enhance energy efficiency.
- Universal Design: The building is accessible to physically challenged and elderly individuals with special parking and entry provisions.
- Cultural Connection: The design ethos draws from the ancient Indian Pancha Bhutas belief, balancing the five elements and integrating local architectural traditions like courtyards and jaalis for climate responsiveness.

Overall, the CII-GBC's design interventions harmonize state-of-the-art technology with traditional Indian architecture to create an eco-friendly, energy-efficient, and culturally resonant commercial building. This approach reduces the ecological footprint while promoting occupant comfort and sustainability leadership.

If more detailed information on specific interventions or materials is required, that can be provided as well.

4.5 Energy modeling

Energy modeling is a computer-based simulation process that replicates a building’s energy use to predict future energy consumption and efficiency. It takes into account factors such as building geometry, materials, location, climate, HVAC systems, lighting, occupancy, and operational schedules. The model combines this data with local weather information to simulate thermal loads, energy use, and occupant comfort over a defined period, often an entire year on an hourly basis.

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Purpose and Benefits

- Helps architects, engineers, and builders evaluate and optimize a building's energy performance during design and operation phases.

- Predicts annual energy consumption, CO2 emissions, daylighting, and solar shading impacts.

- Facilitates testing different scenarios to identify the most energy-efficient systems and design choices before actual implementation.

- Supports compliance with building codes, green building certifications (like LEED), and energy-saving incentives.

- Can be used for building performance rating, HVAC system design, and real-time energy management.

Applications

- Informs trade-offs between construction costs and operational energy savings in design.

- Helps mechanical engineers design HVAC systems tailored to actual building thermal loads.

- Enables large-scale energy efficiency planning for building stock and informs policy and codes.

- Promotes continuous energy-efficient improvements throughout the building's lifecycle.

Energy modeling is thus essential for creating buildings that are energy efficient, cost-effective, environmentally sustainable, and comfortable for occupants[1][2][4][6].

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4.6 HVAC system design

HVAC system design for energy-efficient buildings focuses on creating heating, ventilation, and air conditioning systems that effectively maintain indoor comfort and air quality while minimizing energy consumption and operational costs.

Key Principles of Energy-Efficient HVAC Design

- *Right-Sized Equipment:* Properly sizing HVAC components based on accurate load calculations to avoid energy waste from oversized or undersized units.

- ***High-Efficiency Equipment**:* Using equipment with high energy efficiency ratings like SEER for cooling and AFUE for heating; energy-star rated appliances are preferred.

- *Optimal Layout and Zoning:* Designing HVAC layout with zoning controls to adjust temperatures in different building areas independently, optimizing energy use based on occupancy and usage.

- ***Insulation and Air Sealing**:* Ensuring the building envelope is well insulated and sealed to reduce heat loss or gain, reducing HVAC workload.

- ***Ventilation with Energy Recovery**:* Incorporating energy recovery ventilators (ERV) or heat recovery ventilators (HRV) to pre-condition incoming air and recover energy from exhaust air.

- ***Smart Controls and Automation:*** Utilizing programmable thermostats and building automation systems to optimize HVAC operation schedules based on actual occupancy and time-of-day.

- ***Integration with Building Envelope:*** Aligning HVAC system design with high-performance windows, shading devices, and proper insulation to reduce external heat loads.

- ***Use of Renewable Energy:*** Incorporating renewable energy sources like solar panels or geothermal systems to power HVAC operations where feasible.

- ***Environmentally Friendly Refrigerants:*** Selecting refrigerants with low global warming potential to reduce environmental impact.

- ***Regular Maintenance and Commissioning:*** Ensuring well-maintained systems operate efficiently and commissioning confirms the system meets design specifications.

Benefits

- Efficient HVAC systems reduce energy consumption and greenhouse gas emissions.

- They maintain superior indoor air quality, thermal comfort, and occupant health.

- Proper design extends system life, lowers operational costs, and supports green building certifications.

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This comprehensive approach to HVAC design balances sustainability, cost-effectiveness, and occupant comfort to support long-term building performance and environmental responsibility[1][2][3][5].

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4.7 Chiller Selection

Chiller and pump selection are critical steps in designing efficient HVAC and industrial cooling systems. Both require careful evaluation of system requirements and operating conditions to ensure optimal performance and energy efficiency.

- The most important factor is determining the total cooling load, which is calculated using the formula: $Q=\dot{m}\cdot Cp\cdot \Delta T$ where Q is the heat load, $\dot{m}$ is the mass flow rate, Cp is the specific heat capacity, and ΔT is the temperature difference.
- Key criteria include the design cooling capacity, efficiency (EER & COP), ambient temperature and humidity, refrigerant choice for environmental impact, maintenance access, and future expansion needs.

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- Additional considerations are whether the system should be air-cooled or water-cooled (based on climate, location, and water availability), installation space, maintenance requirements, noise, and lifecycle cost.
- Proper chiller sizing avoids undercooling or over-sizing issues, which can lead to inefficiencies and increased costs.

4.8 Pump Selection

- The primary factors are flow rate (volume of fluid to be circulated), head pressure (the pressure required to overcome system resistance), and compatibility with the fluid properties (viscosity, temperature, chemical composition, etc.).
- Accurate calculation of required flow rate is necessary to ensure adequate heat removal and efficient operation; undersized pumps will not circulate enough fluid, while oversized ones waste energy.
- Head pressure depends on piping layout, elevation changes, fittings, valves, and equipment to ensure pumps can overcome all system resistance.
- Selecting pumps near their best efficiency point (BEP) minimizes energy consumption and operating costs.
- Material compatibility with the fluid (e.g., for corrosive/chilled liquids), maintenance requirements, and total lifecycle costs (purchase, operation, and maintenance) must be considered.

Comparison Table

Selection Aspect	**Chiller**	**Pump**
Sizing Method	Cooling Load Calculation	Flow Rate & Head Pressure
Efficiency Focus	EER, COP	BEP, Variable Speed
Environment Constraints	Air/Water-cooled, Space	Piping Layout, Fluid Type
Fluid Compatibility	Refrigerant, Water	Viscosity, Chemical Makeup
Maintenance	Access, Lifecycle	Reliability, Cost

Proper selection of chillers and pumps ensures energy efficiency, reliable operation, and optimal performance in HVAC and process cooling systems.

Selection of cooling towers depends on several key factors to ensure efficiency, cost-effectiveness, and suitability for the specific application, particularly in commercial or industrial environments.

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4.9 Selection criteria for Cooling Towers

A cooling tower is a device used primarily in industrial and HVAC systems to remove excess heat from water by transferring it to the atmosphere through cooling. It works as a heat exchanger where warm water from processes like air conditioning or industrial operations is cooled down by exposing it to air, causing some water to evaporate, which removes heat and lowers the water temperature.

Selection Criteria for Cooling Towers

- **Cooling Load:** The primary step is to calculate the heat load or the amount of heat that needs to be dissipated, often expressed in BTU/hr or tons of cooling. This is based on hot water flow rate (in GPM) and the temperature difference (range) between inlet and outlet water temperatures.berg-group+1
- **Water Flow Rate and Temperature Requirements:** Accurate knowledge of the circulating water volume and temperature reduction needed determines the tower's capacity and size.deltacooling+1
- **Wet Bulb Temperature:** The ambient wet bulb temperature (the lowest temperature air can be cooled by evaporation) influences cooling tower performance and sizing because it affects cooling efficiency.trane+2
- **Range and Approach:** Range is the temperature difference between hot water entering and cold water leaving the tower. Approach is the difference between cooled water temperature and ambient wet bulb temperature. These factors impact tower design and performance.
- **Cooling Tower Type:** Evaporative (wet) towers cool by water evaporation and are very efficient but use more water, whereas dry cooling towers use air, consume less water, but with lower efficiency. Selection depends on water availability, environmental regulations, and cost considerations.
- **Material Selection and Corrosion Resistance:** Materials like fiberglass reinforced plastic (FRP), galvanized steel, stainless steel, or reinforced concrete are selected based on environmental conditions (humidity, chemical exposure) and maintenance concerns to extend life and reduce repair frequency.
- **Energy Efficiency:** Opt for cooling towers with energy-efficient features such as variable-speed fans and high-performance fill media to reduce operational costs.
- **Noise Levels:** Cooling towers should be selected considering their noise levels, especially in urban or noise-sensitive locations.
- **Maintenance and Durability:** Ease of maintenance, robustness against weather and corrosion, and longevity influence tower type and material choice
- **Cost:** Both initial installation and operational costs (energy, water consumption, maintenance) are vital considerations for long-term sustainability.

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Basic Cooling Load Calculation Formula

Heat Load (British thermal unit)(BTU/hr)=GPM×500×ΔT\text{Heat Load (BTU/hr)} = \text{GPM} \times 500 \times \Delta THeat Load (BTU/hr)=GPM×500×ΔT

Where:

- GPM = water flow rate,(gallons per minute)
- ΔT\Delta TΔT = temperature difference (range) of cooling water,
- 500 = a constant for water (density × specific heat × minutes in an hour).

After determining the nominal load, correction factors based on actual conditions (wet bulb temperature and approach) adjust the final cooling tower capacity required.

Summary Table of Cooling Tower Types

Type	Cooling Method	Water Use	Efficiency	Application Suitability
Wet (Evaporative)	Evaporation cooling	High	High	Most industrial/commercial uses
Dry	Air cooling	None	Lower	Water-scarce locations

Selecting the right cooling tower involves evaluating these factors to match the facility's cooling needs, environmental conditions, and economic goals comprehensively.

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4.10 Selection of air handling units

Selection of air handling units (AHUs) for commercial buildings involves considering several important criteria to ensure optimal indoor air quality, energy efficiency, and space comfort.

An Air Handling Unit (AHU) is a key component of an HVAC system that conditions and circulates air within a building. It is a large indoor unit often consisting of several parts assembled in a frame to handle air processing for heating, cooling, ventilation, and air quality

Key Factors for AHU Selection

- **Building Size and Layout:** The AHU must be sized appropriately to handle the heating, cooling, and ventilation needs of the building space. This depends on the total area, number of floors, zones, and ceiling height.
- **Airflow Requirements:** Airflow capacity (measured in liters per second or cubic feet per minute) is determined based on occupant density, equipment heat loads, and the required air changes per hour for the specific space.
- **Filtration Needs:** The AHU should have filters capable of removing particulates and contaminants present in the environment. The filtration level depends on the building usage—for example, hospitals require high-efficiency HEPA filters, while office spaces may need moderate filtration.
- **Energy Efficiency:** Look for AHUs with high Coefficient of Performance (COP) and Energy Efficiency Ratio (EER) ratings. Features such as variable speed fans, heat recovery systems, and energy-efficient motors help reduce operating costs
- **Material and Construction:** Choose casing and internal components that resist local environmental exposure such as humidity, corrosive air, or outdoor installation. Proper thermal insulation of the casing also reduces energy losses.
- **Maintenance and Accessibility:** An AHU design should allow easy access to filters, coils, and fans for inspection and servicing, minimizing downtime and maintenance costs.
- **Climate and Location:** The local climate affects AHU design details like casing insulation, UV resistance, and corrosion precautions, especially for outdoor units.
- **Customization and Modular Design:** Modular AHUs facilitate easier transport, installation, and adaptation for future building modifications. Customizable units allow specification of coils, filters, controls, and fans tailored to the project needs.

Additional Considerations

- Compliance with local codes and standards such as ASHRAE or Eurovent is necessary for safety, energy efficiency, and IAQ assurance.
- Coordination with building automation systems for smart control adds operational benefits and efficiency.

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Selecting the right AHU is a balance of technical performance, operational cost, environmental conditions, and maintenance practicality, ensuring comfort and reliable air quality in commercial buildings.

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4.11 Precooling of fresh air

in HVAC systems refers to the process of cooling outside air before it enters the main air handling or cooling system. This reduces the load on the HVAC equipment, improves cooling efficiency, lowers energy consumption, and extends equipment lifespan.

How Precooling Works

- **Evaporative Precooling:** Outside air passes through wet pads or cooling media where water evaporates, absorbing heat and lowering the air temperature before it enters the HVAC system. This method is energy-efficient and particularly effective in hot, dry climates without increasing indoor humidity substantially.
- **Adiabatic Cooling:** Uses atomized water mist sprayed into the air stream, cooling the air by evaporation just before the condenser units, reducing intake air temperature by 10-20°C and increasing system capacity.
- **Indirect Evaporative Cooling:** Air is cooled by passing alongside a wetted surface without moisture exchange, producing cooler air temperatures and improved efficiency without adding humidity.

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- **Heat Exchanger Based Systems:** Some systems use heat exchangers to precool air by transferring heat to outgoing warm water or air, enhancing sensible cooling before evaporative processes.

Benefits of Fresh Air Precooling

- **Reduced Energy Costs:** By lowering the temperature of intake air, HVAC systems consume less power for cooling and maintain peak performance, hence reducing operational costs substantially.
- **Improved Equipment Longevity:** Reduced thermal stress on compressors, chillers, and condensers cuts down maintenance needs and prevents premature failures.
- **Increased Cooling Capacity:** Enhanced heat transfer capability allows HVAC systems to better meet cooling demand during peak and extreme temperatures.
- **Stable Indoor Climate:** Precooling helps maintain consistent indoor temperatures essential for occupant comfort and protection of temperature-sensitive processes or products.

Applications

Such precooling technologies are used in commercial buildings, data centers, manufacturing plants, and other facilities where cooling reliability and energy efficiency are critical.

Precooling fresh air is an increasingly popular energy-saving technique to optimize HVAC performance, reduce environmental impact, and lower utility bills.seeleyinternational+2

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4.12 Interior lighting systems

In **green buildings**, the **interior lighting system** is designed to minimize energy consumption, enhance occupant comfort, and reduce environmental impact while maintaining optimal illumination levels. It integrates **natural daylight**, **efficient lighting fixtures**, and **smart controls** to achieve sustainability.

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1. Daylighting

- **Maximizes natural light** through windows, skylights, and light shelves.
- Reduces dependence on artificial lighting during the day.
- **Benefits:**
    - Lowers energy use.
    - Improves occupant mood and productivity.
- **Design considerations:**
    - Proper window orientation (north/south preferred).
    - Use of light shelves and reflective surfaces.
    - Avoid glare and overheating with shading devices.

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2. Energy-Efficient Artificial Lighting

- **LED lights**: Most common; use less energy and have long lifespan.
- **CFLs and OLEDs**: Used where suitable for aesthetics and task lighting.
- **Luminous efficacy**: Focus on fixtures that provide more lumens per watt.

Examples:

- LED panels for general lighting.
- Task lighting with focused LEDs for work areas.
- Motion-sensor-based lights in corridors and restrooms.

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3. Lighting Controls and Automation

Smart systems that adjust lighting based on occupancy and daylight availability.

Common control systems:

- **Occupancy sensors** – Turn lights off when no one is in the room.
- **Daylight sensors** – Dim artificial lights when enough daylight is present.
- **Timers** – Schedule lighting for working hours only.
- **Dimming controls** – Allow users to set preferred light levels.

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4. Light Quality and Human Comfort

- Maintain **balanced color temperature (3000K–5000K)** for visual comfort.

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- Ensure **adequate illuminance** (lux level) per space use:
    - Offices: ~300–500 lux
    - Corridors: ~100–200 lux
    - Classrooms: ~300 lux
- Use **indirect lighting** to reduce glare and eye strain.

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5. Design Integration

- Combine **architectural design** and **lighting design** early in planning.
- Use **highly reflective interior finishes** to enhance brightness.
- Employ **zoning** (separate control for different areas) for flexibility.

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6. Renewable Energy Integration

- Power lighting systems with **on-site solar panels** when possible.
- Use **DC-based LED systems** directly powered by solar PV for high efficiency.

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7. Standards and Certifications

Green buildings follow lighting standards set by:

- LEED (Leadership in Energy and Environmental Design)
- IGBC (Indian Green Building Council)
- BEE (Bureau of Energy Efficiency) – Star ratings for fixtures
- ASHRAE 90.1 – Energy standard for buildings

Key Benefits

- Up to 50–70% reduction in lighting energy consumption
- Lower CO_2 emissions
- Improved indoor environmental quality (IEQ)
- Reduced maintenance costs

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4.13 Features of building

1. **LEED Platinum / Green Rating**
    - First LEED Platinum certified building in India (New Construction v2.0) and one of the earliest outside the USA.
    - More recently, certified as a "Net-Zero Energy" Platinum rated building under the IGBC (Indian Green Building Council).
2. **Energy Efficiency & Renewable Energy**
    - The building uses solar photovoltaic (PV) panels on the roof, satisfying about 20-25% of its energy needs.
    - Passive cooling techniques (e.g. wind towers, evaporative cooling via water pools, courtyards) reduce dependence on mechanical cooling.
3. **Natural Lighting & Interior Comfort**
    - About 90% of interior spaces receive daylight, minimizing artificial lighting until late in the day.
    - Use of high-performance glazing (double glazing with argon gas, etc.), Jali (perforated screen walls) to control glare and heat while allowing light and ventilation.
4. **Water Conservation and Management**
    - Zero or almost zero wastewater discharge: all grey & black water is treated via Root Zone Treatment System, biologically, then reused (e.g. for landscaping).
    - Rainwater harvesting, permeable surfaces to enable storm water seepage.
5. **Green Roofs / Vegetation**
    - A large part of roof area (≈ 55-60%) is covered with vegetation (roof gardens) which help insulation, reduce urban heat island effect.
    - Landscaping using native/adaptive trees, retaining as much of the site's natural topography and rock structures as possible.
6. **Use of Recycled / Local Materials**
    - Many building materials are recycled or recyclable (fly-ash bricks, recycled glass, aluminium, ceramic tiles, bagasse boards instead of plywood, etc.).
    - Sourcing materials locally (within ~500 miles) to reduce transportation impact.
7. **Building Management / Monitoring**
    - Real-time monitoring of energy use, water use, indoor air quality via Building Management System (BMS).

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- Use of low Volatile Organic Compounds (VOC) paints, adhesives etc. to improve indoor environment.

8. Design & Orientation

- Building orientation via simulation to minimize heat gain but maximize daylight.
- Active/passive design features like courtyards, wind towers, shading devices.

9. Water and Energy Savings

- Around 40-35% reduction in potable water use versus conventional buildings.
- Overall energy savings (including lighting, HVAC) estimated at ~50% reduction relative to comparable buildings.

10. Demonstration & Advocacy Role

- The centre is not just a green building but a demonstration/living lab, showing green building techniques, advising industries, architects, policy makers.
- It helps spread green practices across India.

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Unit 5

Material Conservation

5.1 Handling of non process waste

The handling of **non-process waste in green buildings** focuses on reducing, reusing, and recycling materials to minimize waste sent to landfills and decrease environmental impact. Non-process waste refers to solid waste not directly tied to a building's manufacturing or operational processes—such as office waste, packaging, food waste, and discarded materials from daily activities.

Core Principles of Non-Process Waste Management

Green buildings adopt the **3R approach (Reduce, Reuse, Recycle)** to manage non-process waste efficiently. The Indian Green Building Council's (IGBC) *Net Zero Waste Rating System* and LEED (Leadership in Energy and Environmental Design) both emphasize this method.

1. **Reduction**
    - Designing with durable and minimal packaging materials.
    - Encouraging digital documentation to minimize paper use.
    - Using reusable containers and refillable supplies.
2. **Reuse and Repurposing**
    - Repurposing discarded materials such as pallets or glass for decorative or furniture use.
    - Donating surplus furniture and equipment to local organizations.
3. **Recycling and Composting**
    - Providing clearly labeled recycling bins for paper, plastics, metals, glass, and e-waste.
    - Composting food and organic waste to create soil enrichers for landscaping or rooftop gardens.

Design and Operational Strategies

Non-process waste management begins at the **design stage** and continues through **operation**:

- **Design Phase:** Integrate a solid waste management plan into building specifications and allocate space for waste segregation and storage facilities.
- **Construction & Occupancy:** Apply a waste tracking system to monitor quantities and types of waste generated, ensuring accountability and optimization.
- **Smart Waste Systems:** Buildings increasingly use **digital waste management systems** that track waste levels, automate sorting, and provide real-time data for audits and compliance.

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Certification and Policy Frameworks

Green building certification systems, such as **LEED**, **GRIHA**, and **IGBC**, provide credit points for integrated waste management practices:

Framework	Key Requirement	Focus Area
LEED	Dedicated area for recyclable storage and regular waste audits	Recyclables segregation
GRIHA	Waste-to-resource strategies like composting or reuse	Sustainable disposal
IGBC Net Zero Waste	Equilibrium between waste generated and recycled; aim for zero landfill waste	Waste diversion and resource recovery

Sustainable Outcomes

Effective handling of non-process waste in green buildings:

- Minimizes landfill use and pollutant generation.
- Reduces carbon footprint through waste diversion.
- Enhances compliance with sustainability goals and green certifications.
- Lowers operational costs by reducing disposal expenses and reusing materials.

In essence, sustainable non-process waste handling integrates design foresight, occupant participation, and advanced monitoring technologies to foster circular resource use in green building ecosystems.

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5.2 Waste reduction during construction

Waste reduction during construction is a central principle of sustainable building practice, aiming to minimize environmental impact, reduce costs, and improve material efficiency. It focuses on preventing waste generation rather than managing it after production through strategic planning, technology adoption, and resource optimization.

Key Strategies for Reducing Waste

1. **Smart Planning and Design**
   Early-stage planning can prevent up to 50% of potential construction waste. Using **Building Information Modeling (BIM)** allows designers to optimize material use, avoid clashes, and predict waste streams before construction begins. Designing for **modularity, flexibility, and deconstruction** ensures materials can be disassembled and reused at the end of the building's life.

2. **Accurate Material Estimation and Procurement**
   Ordering precise material quantities minimizes over-purchasing and leftover materials. Collaborating with suppliers who offer return programs, bulk packaging reduction, or reusable delivery systems supports waste reduction efforts. Adopting **just-in-time inventory systems** ensures that materials arrive when needed, reducing losses due to damage, theft, or exposure.

3. **Use of Sustainable and Recycled Materials**
   Incorporating materials such as **recycled concrete aggregate, fly ash bricks, reclaimed steel, and bamboo** reduces the demand for virgin resources and lowers waste generation. Reusing salvaged materials from demolition also supports a circular economy approach.

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4. **Lean Construction Principles**
   Lean construction focuses on eliminating operational inefficiencies. Practices like **Value Stream Mapping** and the **5S methodology** (Sort, Set, Shine, Standardize, Sustain) improve site organization and reduce defects or rework that produce waste. This streamlining of workflows contributes significantly to resource conservation.

5. **Prefabrication and Modular Building**
   By manufacturing building components in a factory-controlled environment, **prefabrication** reduces cutting waste, packaging materials, and on-site errors. It also enhances quality control and speeds up construction schedules.

6. **Recycling, Reuse, and Upcycling**
   Establishing systems for **segregation of construction waste** ensures recycling opportunities. Recycled concrete can serve as road sub-base, while metals and timber can be repurposed. **Upcycling**—turning waste materials into products of higher value—is increasingly being used for design finishes and landscape features.

7. **Training and Accountability**
   Educating construction workers about sustainable practices and implementing site-level **waste audits** help maintain accountability and continuous improvement. Digital tools such as **Project Management Information Systems (PMIS)** monitor waste generation in real time, providing data for adaptive planning.

Benefits of Waste Reduction

Implementing effective waste reduction during construction results in:

- Reduced landfill burden and greenhouse gas emissions.
- Lowered project costs through efficient resource use.
- Improved compliance with green building certifications like LEED, IGBC, and GRIHA.
- Enhanced sustainability branding and reputation for construction firms.sbnsoftware+1

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5.3 Materials with recycled content

Materials with recycled content are widely used in green building design to reduce resource extraction, minimize landfill waste, and lower embodied carbon. These materials are manufactured using post-consumer or post-industrial waste as a raw input while maintaining high structural and aesthetic performance.

Common Materials with Recycled Content

1. **Recycled Steel**
   Structural steel is the most recycled construction material, with nearly 98% of steel recycled globally. It retains its strength after multiple recycling cycles and is commonly used for beams, rebar, framing, roofing, and façade systems.

2. **Recycled Concrete Aggregate (RCA)**
   Old concrete from demolition or road projects is crushed and reprocessed for new construction use. RCA replaces natural aggregates in sub-base layers, pavement, and foundation fills, reducing quarrying impact and embodied energy.

3. **Reclaimed Wood**
   Salvaged timber from old structures or pallets is reused for flooring, cladding, furniture, and framing. It offers natural aesthetics, reduces deforestation, and

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contributes to LEED and IGBC credits under recycled and regional material categories.

4. **Recycled Plastic**
   Plastic waste from bottles and industrial products is converted into durable materials such as composite lumber, bricks, panels, roof tiles, and floor tiles. These materials are lightweight, water-resistant, and reduce landfill plastic accumulation.

5. **Recycled Glass**
   Glass cullet is remelted and used in countertops, tiles, terrazzo, insulation, and concrete aggregate (glassphalt). It is visually appealing, non-toxic, and can contain up to 100% recycled content without loss of performance.

6. **Recycled Gypsum (Plasterboard)**
   Made by reprocessing gypsum waste from drywall, recycled plasterboard can achieve up to 96% recycled content and reduces the need for virgin gypsum mining.

7. **Recycled Rubber**
   Derived from end-of-life tires, this material is used in flooring systems for sports facilities, playgrounds, and acoustic insulation. It provides comfort, durability, and slip resistance.

8. **Recycled Paper and Cellulose Insulation**
   Recycled paper fibers form cellulose insulation, an alternative to fiberglass. Treated for fire and pest resistance, it has excellent thermal and acoustic performance while avoiding the need for new pulp processing.

Benefits and Performance

- **Energy and Resource Efficiency:** Using recycled content reduces raw material extraction and energy intensity in manufacturing.
- **Lower Embodied Carbon:** Recycling processes require less energy compared to producing virgin materials.
- **Circular Economy Support:** Promotes reuse cycles within construction and manufacturing streams.
- **Green Certification Credits:** Systems like LEED, IGBC, and GRIHA award points for materials with high recycled content.

Application Areas in Green Buildings

Material	Common Uses	Typical Recycled Content (%)*
Recycled Steel	Structural frames, facades	25–98
Recycled Concrete	Aggregate, pavement	30–90
Recycled Plastic	Panels, tiles, bricks	50–100
Recycled Glass	Countertops, insulation	27–100

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Material	Common Uses	Typical Recycled Content (%)*
Recycled Gypsum	Drywall, ceiling boards	70–96
Reclaimed Wood	Flooring, joinery	Variable

(*Typical recycled content values vary based on sourcing and process.)

Overall, integrating materials with recycled content supports the transition toward a circular, low-impact construction ecosystem that enhances both building resilience and resource sustainability.

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5.4 Local materials

Local materials in construction refer to naturally available or regionally manufactured resources sourced close to the building site. Their use minimizes transportation energy, supports the local economy, and aligns with sustainable construction principles that lower embodied carbon and environmental impact.

Importance of Using Local Materials

Using local materials promotes resource efficiency and reduces carbon footprints by cutting down long-distance transportation and reliance on imported products. It also supports local industries, enhances cultural relevance, and boosts building adaptability to the climate and soil conditions of the region.

Common Examples of Local Construction Materials

1. **Bamboo**
   Bamboo is a rapidly renewable material found abundantly in Asia, including India. It has high tensile strength, making it suitable for flooring, scaffolding, roofing, and even structural frames in eco-friendly construction.

2. **Rammed Earth**
   Created using compacted local soil (sand, clay, and gravel), rammed earth walls are durable, thermally efficient, and completely recyclable. This ancient technique blends cultural heritage with modern sustainability practices.

3. **Fly Ash Bricks**
   Made from byproducts of coal-fired plants, these bricks are widely available across India. They have superior strength, thermal insulation, and lower embodied energy than traditional clay bricks.

4. **Natural Stone**
   Stones like laterite, slate, and granite are locally quarried and often used for walls, flooring, and facades. They offer high durability, minimal maintenance, and can be sourced without heavy industrial processing.

5. **Clay and Terracotta**
   Locally extracted clay is used for making terracotta tiles, earthen blocks, and pottery-based architectural elements. These materials are biodegradable, breathable, and contribute to indoor thermal comfort.

6. **Lime Mortar and Pozzolanic Additives**
   Traditional lime mortars made from locally available limestone offer low energy processing compared to cement. Combining lime with pozzolanic materials, such as volcanic ash or fly ash, increases strength and sustainability.

7. **Timber and Reclaimed Wood**
   Locally harvested or reclaimed timber reduces deforestation pressure when managed

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under sustainable forestry practices. It is commonly used for joinery, furniture, and flooring.

Benefits of Using Local Materials

- Reduced Transportation Energy: Decreases fossil fuel consumption and related greenhouse gas emissions.
- Economic Support: Encourages regional industries and crafts-based employment.
- Cultural Integration: Reflects the vernacular architecture and traditions of the region.
- Improved Thermal Comfort: Materials such as clay, bamboo, and stone adapt naturally to local climate conditions.
- Enhanced Sustainability: Minimizes waste generation and promotes resource circularity.

In certification systems such as LEED, IGBC, and GRIHA, sourcing materials within a specific radius (usually 400–800 km) earns credits under *Regional Material* or *Sustainable Building Material* categories. These systems encourage architects to utilize locally available materials while ensuring they are non-toxic, renewable, and durable

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5.5 Materials reuse

Material reuse in construction refers to recovering and reapplying existing building elements or materials—either directly or after minor reprocessing—to reduce waste, conserve resources, and support a circular economy. It is a cornerstone of sustainable construction, minimizing the need for new material production and lowering embodied energy.

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Key Strategies for Material Reuse

1. **Deconstruction Instead of Demolition**
   Deconstruction is the deliberate disassembly of buildings to salvage usable components such as bricks, wood, doors, and fixtures rather than destroying them. This method maximizes material recovery, reduces landfill waste, and creates employment opportunities in material reclamation and reprocessing.

2. **Designing for Disassembly and Adaptability**
   Buildings designed with modular systems, accessible connections (bolts and screws rather than adhesives), and durable components can be dismantled and reused in future projects. This approach extends a building's life and permits cost-effective adaptation to new uses.

3. **Material Exchanges and Salvage Centers**
   Reuse networks and material banks connect contractors, designers, and individuals to trade reclaimed materials. Salvage centers facilitate the resale of structural and architectural components like flooring, fittings, or façade materials, supporting local circular economies.

4. **On-Site Reuse and Repurposing**
   Many materials can be reused on construction sites with little processing:

   - Concrete and masonry can be crushed into aggregate for sub-base or road construction.
   - Wood offcuts serve as formwork or blocking.
   - Insulation waste can be reused for internal walls.
   - Paint can be remixed as undercoats or used in garages and service areas.

5. **Upcycling**
   Upcycling transforms waste into new, higher-value products. Examples include using reclaimed wood for designer furniture, repurposing old windows as interior partitions, or turning steel beams into decorative stair structures. This reduces landfill waste while adding bespoke aesthetic value.

Materials Commonly Reused

Material	Common Reuse Applications	Benefits
Concrete & Masonry	Crushed for sub-base, flooring blocks	Reduces quarrying
Wood	Refurbished furniture, joinery, formwork	Conserves forest resources
Steel	Structural reuse, fencing, reinforcement	100% recyclable with no strength loss
Brick	Facades, paving, decorative	Retains embodied energy

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Material	Common Reuse Applications	Benefits
	walls	
Glass	Decorative panels, windows, tiles	Low energy processing
Fixtures & Fittings	Lighting, doors, HVAC grilles	Reduces procurement cost

Environmental and Economic Benefits

- Reduced Carbon Footprint: Avoids emissions from manufacturing and transport of new materials.
- Resource Conservation: Preserves natural materials like timber, metals, and aggregates.
- Lower Costs: Minimizes material procurement and waste disposal expenses.
- Cultural Preservation: Maintains architectural elements from heritage structures through adaptive reuse.

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5.6 Certified wood

Certified wood refers to wood and wood products that come from forests managed according to strict environmental, social, and economic standards verified by independent third-party organizations. These certifications ensure sustainable forest management, protecting biodiversity, preventing deforestation, and supporting community rights.

Key Certification Programs for Wood

- **Forest Stewardship Council (FSC)**
  FSC certification is a globally recognized standard ensuring that wood is sourced from responsibly managed forests with ecological conservation, social responsibility, and economic sustainability. The FSC label guarantees wood traceability through the supply chain from forest to end product.
- **Program for the Endorsement of Forest Certification (PEFC)**
  PEFC is an international non-profit that endorses national forest certification systems. PEFC-certified wood ensures forests are managed sustainably according to strict local and international requirements and promotes supply chain transparency with chain of custody certification for construction projects.
- **Sustainable Forestry Initiative (SFI)**
  SFI is a North American forest certification standard supporting sustainable forestry practices. It emphasizes forest health, biodiversity, and community benefits and is recognized by many green building programs.

Importance in Green Building

- Using certified wood contributes to environmentally responsible construction by promoting sustainable forestry and reducing illegal logging.
- Certified wood products help projects earn green building credits, such as those awarded by the Indian Green Building Council (IGBC) when at least 50% of wood-based materials come from FSC-certified sources.
- Certification supports chain of custody integrity, enabling builders and architects to verify wood sourcing throughout procurement and construction phases.

Common Certified Wood Products

- Structural framing and beams
- Flooring and paneling
- Engineered wood products like glulam, cross-laminated timber (CLT), and laminated veneer lumber (LVL)
- Doors, windows, and cabinetry
- Temporary construction uses like concrete formwork and pedestrian barrierspefc+1

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5.7 Rapidly renewable building materials and furniture

Rapidly renewable building materials and furniture are sourced from plants or crops that regenerate quickly—usually within a 10-year cycle or less—making them highly sustainable alternatives to traditional slow-growing timber and resource-intensive materials. Their rapid growth rates help reduce environmental impact by ensuring a steady supply and lowering pressure on forests.

Common Rapidly Renewable Building Materials

1. **Bamboo**
   Bamboo is among the fastest-growing plants, maturing in 3 to 5 years. It is widely used in flooring, wall panels, structural elements, veneers, and scaffolding due to its strength, flexibility, and aesthetic appeal.
2. **Cork**
   Harvested from the bark of cork oak trees without cutting them down, cork regenerates every 9 years. It is used for flooring, wall coverings, soundproofing, and insulation. Cork's natural elasticity and durability make it ideal for furniture surfaces as well.
3. **Wheatboard and Strawboard**
   Compressed wheat straw or agricultural residues form panels used as sustainable substitutes for plywood or MDF. These materials are biodegradable, lightweight, and have lower embodied energy.
4. **Palm Wood**
   Derived from the trunks of palm trees, palm wood is an alternative to hardwood and is

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used in flooring, furniture, and decorative elements. Palm trees grow and mature in less than 10 years.

5. **Rattan**
   Rattan, a climbing palm, grows faster than hardwood and is widely used for furniture making due to its light weight and flexibility. It is also a renewable option supporting local economies in tropical regions.

6. **Fast-Growing Softwoods**
   Species like pine and fir, which mature quickly (within 15 years), are increasingly used in furniture and framing to supplement slower-growing hardwoods.

Benefits in Building and Furniture

- Environmental Impact: Rapid renewability means these materials sequester carbon more quickly and reduce deforestation pressures.
- Economic and Social: Often cultivated in agroforestry or managed farms, supporting rural livelihoods and sustainable land use.
- Durability and Adaptability: Innovations in treatment and manufacturing improve longevity and resistance to pests and moisture.
- Aesthetic Appeal: Natural textures and finishes are popular in modern sustainable design.

Applications

Material	Typical Uses	Renewal Cycle (Years)
Bamboo	Flooring, structural beams, panels, furniture	3-5
Cork	Flooring, wall tiles, furniture surfaces	9
Wheatboard/Strawboard	Cabinets, partitions, panels	<1 (annual crop)
Palm Wood	Flooring, furniture, trim	<10
Rattan	Furniture, baskets, decorative elements	5-7

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Material	Typical Uses	Renewal Cycle (Years)
Fast-Growing Softwoods	Framing, furniture, moldings	10-15

In summary, rapidly renewable materials in building and furniture offer a practical, eco-friendly path to minimize the environmental footprint while supporting sustainable forestry and agriculture. Their increasing adoption aligns with global green building goals and circular economy principles.

5.8 Indoor environment quality and occupational health

Indoor environmental quality (IEQ) in green buildings is generally improved compared to conventional buildings, leading to measurable benefits in occupant health and comfort. Green buildings typically exhibit lower levels of indoor air pollutants such as volatile organic compounds (VOCs), formaldehyde, allergens, environmental tobacco smoke (ETS), nitrogen dioxide (NO2), and particulate matter (PM), which are common contributors to poor health outcomes. These improvements arise from sustainable design choices like better ventilation systems, the use of low-emission materials, and overall healthier indoor environments. Occupants of green buildings report fewer sick building syndrome symptoms, fewer respiratory problems, and experience better physical and mental health. Additionally, improved IEQ translates to better work productivity, lower absenteeism, and overall enhanced well-being in workplaces and other green building environments.

Occupant health in green buildings also involves ensuring worker safety and well-being during the construction and operation phases. While green buildings aim to minimize environmental harm and provide healthier spaces, there are unique safety challenges tied to their construction such as risks of falls and overexertion for workers. These occupational health and safety concerns require careful risk management and safety practices to protect construction workers involved in green building projects. However, once operational, green buildings tend to support healthier working conditions, reducing exposure to pollutants and promoting greater occupant comfort and satisfaction.

Indoor Environmental Quality Benefits in Green Buildings

- Lower concentrations of harmful pollutants (VOCs, PM2.5, formaldehyde)
- Enhanced ventilation systems and use of sustainable materials reduce indoor contamination
- Improved thermal comfort, humidity control, and better air quality consistency
- Fewer reports of symptoms such as headaches, fatigue, and respiratory issues
- Higher occupant satisfaction related to indoor environment parameters such as temperature, humidity, and lighting

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Occupational Health in Green Building Construction and Operation

- Construction presents hazards like falls from heights and overexertion, requiring focused safety management
- Proper risk assessments and safety practices are essential during green building projects
- Operational green buildings promote healthier work environments with fewer health complaints and better productivity
- Green building design elements can contribute to reduced absenteeism and improved employee well-being

5.9 Air conditioning

Air conditioning is the controlled process of regulating air temperature, humidity, purity, and circulation within an enclosed space to maintain a comfortable indoor environment, regardless of external weather conditions. It serves both residential and commercial applications and is a key part of HVAC (heating, ventilation, and air conditioning) systems.

How Air Conditioning Works

Air conditioning systems operate through the vapor-compression refrigeration cycle, which involves the continuous movement and phase change of a refrigerant to transfer heat. The major components are the compressor, condenser coil, expansion valve, and evaporator coil.

1. Evaporator Coil: The refrigerant absorbs heat from indoor air as it evaporates from a liquid to a gas, cooling and dehumidifying the air.
2. Compressor: The gas is pressurized, raising its temperature and pressure.
3. Condenser Coil: The hot gas releases its heat to the outside environment as it condenses back into a liquid.
4. Expansion Valve: The high-pressure liquid passes through this valve, lowering its pressure and temperature before returning to the evaporator to repeat the cycle.

This cycle effectively removes heat from indoors and releases it outdoors, creating a cooled and more comfortable interior climate.

Functions of Air Conditioning

Air conditioning systems manage four fundamental aspects of indoor air quality:

- Temperature Control – Maintaining desired thermal comfort.
- Humidity Control – Reducing excessive moisture levels.
- Airflow Regulation – Ensuring adequate movement and distribution of conditioned air.
- Air Purity – Filtering pollutants and improving air hygiene.

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Types of Air Conditioning Systems

Common forms of air conditioning include:

- Window Units – Compact systems for single rooms.
- Split Systems – Separate indoor and outdoor units, common in homes.
- Packaged Units – Self-contained systems used in buildings.
- Ductless Mini-Split Systems – Flexible systems for spaces without ductwork.
- Chilled-Water Systems – Used in large commercial facilities for centralized cooling.

5.10 Indoor air quality

Indoor air quality (IAQ) refers to the condition of the air inside buildings, particularly regarding its impact on the health, comfort, and performance of occupants. As people spend nearly 90% of their time indoors, IAQ plays a central role in human well-being, productivity, and disease prevention.

Key Determinants of Indoor Air Quality

The main factors influencing IAQ include pollutant levels (e.g., volatile organic compounds or VOCs, particulate matter, carbon dioxide, and carbon monoxide), humidity, temperature, ventilation rate, and the presence of biological contaminants such as mold, pollen, or bacteria. Poor IAQ can lead to symptoms of sick building syndrome, fatigue, headaches, allergic reactions, and long-term respiratory or cardiovascular conditions.

Impact of Green Building Practices on IAQ

Green buildings are specially designed to improve IAQ by integrating sustainable materials, advanced ventilation systems, and natural design principles that minimize pollutant sources.

- Low-emission materials: Green buildings use low- or zero-VOC paints, adhesives, and finishes that prevent harmful chemical release.
- Improved ventilation and filtration: High-efficiency mechanical ventilation and natural airflow systems ensure steady removal of indoor pollutants.
- Indoor greenery: Plants absorb airborne toxins and increase oxygen levels, acting as biological air filters.
- Moisture regulation: Natural materials such as clay maintain stable humidity, limiting mold growth.

These integrated strategies produce measurably better air quality outcomes—lower formaldehyde, nitrogen dioxide, and particulate matter levels, leading to improved physical health and comfort in occupants.

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Health and Cognitive Benefits

Good indoor air quality correlates strongly with better cognitive performance, productivity, and mental well-being. Studies by the World Green Building Council found that occupants in buildings with enhanced ventilation scored nearly 300% higher on cognitive tasks compared to those in conventional environments. High IAQ also reduces the prevalence of respiratory illnesses, allergies, and absenteeism, offering both health and economic benefits for building owners and tenants.

Measuring and Managing IAQ

IAQ is typically assessed through pollutant concentrations (VOCs, CO_2, PM2.5) and ventilation efficiency metrics. Green building frameworks
like LEED and WELL integrate IAQ performance standards, requiring continuous monitoring and maintenance to ensure sustained air purity.

5.11 Sick building syndrome

Sick Building Syndrome (SBS) is a condition where occupants of a building experience a range of acute health and comfort effects that seem linked to time spent in the building, though no specific illness or contaminant can be identified as the cause. The most distinctive feature is that symptoms tend to improve after leaving the building and worsen upon returning.

Main Symptoms

SBS can affect multiple body systems, often mimicking common illnesses such as colds or allergies. Key symptoms include:

- Headaches, dizziness, and difficulty concentrating
- Eye, nose, and throat irritation or dryness
- Fatigue, irritability, and nausea
- Skin dryness or rash
- Respiratory issues including chest tightness, wheezing, and coughing
- Allergy-like symptoms such as sneezing and nasal congestion.

Causes and Contributing Factors

The causes of SBS are multifactorial, often associated with poor indoor air quality and inadequate ventilation. Common contributing factors include:

- Chemical contaminants such as volatile organic compounds (VOCs), formaldehyde, asbestos, carbon monoxide, tobacco smoke, and cleaning products.
- Biological contaminants including mold, fungi, bacteria, and pollen.
- Inadequate ventilation systems, which fail to remove indoor pollutants and dilute indoor air with fresh outdoor air.

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- Poor lighting, high noise levels, and thermal discomfort which contribute to environmental stress.
- Psychosocial factors like high workload or low job satisfaction, which can exacerbate SBS symptoms.

Historically, energy conservation measures reduced outdoor air ventilation rates in buildings after the 1970s oil crisis, which contributed to increased SBS reports in modern, tightly sealed buildings.

At-Risk Groups

Individuals spending extended periods in poorly ventilated spaces, particularly office workers, are most at risk. Other risk factors include having asthma, high chemical sensitivity, or living in newly built or recently renovated buildings where off-gassing materials release pollutants.

Prevention and Control

Effective strategies to prevent or alleviate SBS include:

- Increasing ventilation rates and ensuring proper HVAC maintenance.
- Using low-emission building materials such as low-VOC paints and adhesives.
- Controlling temperature and humidity to limit microbial growth.
- Implementing regular cleaning and maintenance to remove dust and mold.
- Encouraging access to natural light and airflow for occupant well-being.

5.12 Tobacco smoke control

Controlling tobacco smoke indoors is crucial for maintaining good indoor air quality and protecting the health of both smokers and non-smokers exposed to second hand smoke. Tobacco smoke contains thousands of toxic chemicals and carcinogens that significantly degrade indoor air quality, leading to respiratory and cardiovascular problems among other health risks.

Key Methods for Tobacco Smoke Control Indoors

- **Source Control**: The most effective method is to eliminate or restrict smoking indoors entirely. Designating outside-only smoking areas greatly reduces indoor contamination and exposure.
- **Enhanced Ventilation:** Increasing air exchange rates by opening windows and doors or using exhaust fans helps to dilute smoke concentrations. Mechanical ventilation systems like HVAC with outdoor air intake or heat recovery ventilators can further improve air freshness.
- **High-Efficiency Air Filtration:** Installing HEPA filters and activated carbon filters in HVAC systems or standalone air purifiers can capture fine particulate matter and help reduce odors and harmful chemicals from tobacco smoke.

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- **Regular HVAC Maintenance:** Cleaning and maintaining ducts and filters ensures that tobacco smoke particles are effectively trapped and prevents recirculation of polluted air.
- **Use of Air Purifiers:** Portable air purifiers equipped with filtration technologies can be placed in smoking zones to reduce airborne contaminants.

Although ventilation and filtration techniques help reduce indoor tobacco smoke levels, they do not completely eliminate the risks of secondhand and thirdhand smoke exposure, so the best practice remains preventing indoor smoking altogether.

5.13 Minimum fresh air requirement avoid use of asbestos in the building

Minimum fresh air requirements in buildings are guided primarily by standards such as ASHRAE Standard 62.1 (for commercial buildings) and Standard 62.2 (for residential buildings), which specify ventilation rates necessary to ensure acceptable indoor air quality and minimize adverse health effects.

Minimum Fresh Air Requirements

- ASHRAE 62.1 recommends fresh air ventilation rates expressed in cubic feet per minute per person (cfm/person). Typically, a minimum of 5 to 10 cfm of outdoor air per person is required depending on the building type and use, with specific rates varying for different space types, such as offices, retail stores, or hotel guest rooms.
- For residential buildings, ASHRAE 62.2 suggests a minimum ventilation rate of 0.35 air changes per hour or at least 15 cfm per person, ensuring sufficient outdoor air exchange to dilute indoor pollutants.
- Other guidelines such as the CSIR(The council of scientific and industrial research) of recommend a minimum fresh air supply of around 10 liters per second per person (36 m^3/person/hour) as a general baseline for minimizing airborne contaminants and pathogens indoors.

Avoiding Asbestos Use in Buildings

- Asbestos—a hazardous material once widely used for insulation, fireproofing, and soundproofing—is now largely banned or severely restricted in many countries due to its carcinogenic health risks.
- Avoiding asbestos involves selecting alternative, non-toxic building materials that provide similar fire resistance and insulation properties, such as fiberglass, cellulose, foam, or other modern composites.
- Strict regulations mandate that existing asbestos-containing materials (ACMs) be safely managed, removed, or encapsulated during building renovations or demolitions to prevent exposure to airborne asbestos fibers.
- Green building certifications and health-focused standards emphasize the use of safe, sustainable materials free of asbestos and other harmful substances to promote healthier indoor environments.

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5.14 Improved fresh air ventilation

Improved fresh air ventilation in green buildings is achieved through a combination of smart design, adherence to standards like ASHRAE 62.1, natural ventilation strategies, and advanced mechanical systems, all aimed at maximizing occupant health and minimizing energy consumption.

Strategies for Fresh Air Ventilation

- Integrating energy recovery ventilators (ERVs) or heat recovery ventilators (HRVs) enables continuous exchange of stale indoor air with fresh outdoor air, while minimizing energy loss in tightly sealed, energy-efficient buildings.
- Natural ventilation techniques, such as operable windows, skylights, clerestories, and cross-ventilation, help utilize wind and thermal buoyancy to improve air quality and reduce reliance on mechanical systems.
- Displacement ventilation systems distribute fresh air from low-level diffusers, allowing it to rise and exhaust through ceiling vents, thus efficiently diluting contaminants at the breathing zone and enhancing comfort.

Key Standards and Requirements

- The ASHRAE 62.1 standard is widely referenced, specifying minimum fresh air requirements based on occupancy and floor area—for example, 8 liters/second per person, or defined cubic feet per minute (cfm) per person and per unit area, as well as formulas for calculating the required ventilation rates in various building types.
- Indian and international green building certifications, like IGBC and LEED, mandate compliance with these ventilation requirements to ensure adequate air exchange and occupant health.

Benefits of Enhanced Fresh Air Ventilation

- Improved indoor air quality reduces concentrations of indoor pollutants, volatile organic compounds (VOCs), and allergens, preventing sick building syndrome and respiratory problems.
- Enhanced ventilation creates a healthier, more comfortable indoor environment, supporting occupant well-being and productivity.

- Using natural ventilation and air exchange strategies can significantly lower energy consumption by reducing the need for air conditioning and mechanical ventilation, thus reducing operational costs and carbon emissions.
- **Design Considerations**
  - Effective implementation involves site-specific planning (orientation, climate, window placement), proper building envelope insulation, and automated controls for operable openings to balance energy efficiency with adequate air flow.
  - Combining smart controls, filtration (HEPA, CO2-controlled systems), and both passive and active systems further enhances performance in modern green buildings.

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Improved fresh air ventilation is vital for sustainable building performance, offering significant health, comfort, and energy-saving benefits when integrated through thoughtful design and adherence to established standards.

5.15 Measure of IAQ

Indoor air quality (IAQ) in green buildings is typically assessed using quantifiable measures such as the concentration of indoor pollutants, ventilation rates, and the presence of certain environmental parameters that affect occupant health and comfort.pmc.ncbi.nlm.nih+2

Key IAQ Measures in Green Buildings

- **Particulate Matter (PM2.5/PM10):** Daily average concentration is measured, with strict limits—often <37.5 µg/m³ for PM2.5 according to healthy building standards.
- **Carbon Dioxide (CO_2):** CO_2 levels are monitored to ensure proper ventilation and occupant comfort, with typical limits set to 1000 ppm as per standards.
- **Volatile Organic Compounds (VOCs) & Formaldehyde (CH_2O):** VOCs and formaldehyde concentrations are capped, with green building certifications requiring levels well below thresholds (e.g., formaldehyde <30 µg/m³ for aspirational targets).
- **Total Microbial Count:** Microbial count is measured to prevent adverse health effects from bacteria, molds, and other microbes in regularly occupied spaces.
- **Relative Humidity (RH) and Temperature:** Both are monitored and managed to maintain comfort. For example, in cooling season, thermal comfort zone ranges are often 24–26°C and 40–60% RH.
- **Air Filtration Efficiency:** High-efficiency filters are used in HVAC systems to capture particulate and biological pollutants.
- **Ventilation Rates:** Measurement and provision of adequate outdoor air ventilation, based on occupancy and building use.
- **Indoor Air Monitoring Systems:** Real-time IAQ monitoring solutions track multiple parameters to ensure compliance with green building certification (such as LEED, WELL, GRIHA, IGBC).

Standards and Certification Criteria

- Certifications like LEED, WELL, GRIHA, and IGBC require routine IAQ monitoring, documentation, and achievement of thresholds for pollutants and comfort parameters.
- ISHRAE Standard 10001:2016 and ISO 16000 series provide measurement methodologies and instrument specifications for IAQ assessment.
- IAQ credit points are awarded for both meeting thresholds and implementing emissions reduction strategies, such as using low-VOC materials and source control.

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Practical Implementation

- Green buildings employ a combination of source control (low-emission materials), enhanced ventilation, regular monitoring, moisture management, and advanced filtration.
- IAQ data collection involves sensors placed in representative areas, with parameters logged at intervals and compared against standards for compliance and certification.

In summary, measurable IAQ indicators in green buildings include pollutant concentrations, ventilation rates, temperature and humidity, filtration effectiveness, and microbial counts—each supported by standards and real-time monitoring for certification purposes.

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5.16 Reasons for poor IAQ

Poor indoor air quality (IAQ) arises primarily from a combination of pollutant sources, inadequate ventilation, and environmental factors such as humidity. Key reasons include:

- **Indoor pollutant sources:** Combustion appliances (gas stoves, fireplaces), tobacco smoke, building materials (e.g., asbestos, pressed wood releasing formaldehyde), cleaning products, insecticides, and furniture off-gassing release harmful gases and particles into the air.
- **Poor ventilation:** Insufficient fresh air exchange causes contaminants to build up indoors, leading to stale, polluted air. Modern energy-efficient buildings that are tightly sealed without adequate mechanical ventilation can trap pollutants.
- **High humidity:** Excess moisture encourages mold, mildew, dust mites, and microbial growth which degrade IAQ and can provoke allergies or respiratory issues. Humidity also can cause certain materials to emit irritant gases like formaldehyde.
- **Outdoor pollutant infiltration:** Pollutants such as radon, pollen, and vehicle emissions can enter buildings through openings, cracks, and ventilation systems, adversely affecting indoor air.
- **Human activities:** Smoking, cooking (especially with solid fuels), cleaning, and hobbies can intermittently release spikes of pollutants indoors.
- **Improper maintenance:** Malfunctioning or unvented appliances can elevate indoor concentrations of carbon monoxide and other pollutants.

5.17 Measures to achieve acceptable IAQ levels

Measures to achieve acceptable indoor air quality (IAQ) levels focus on controlling pollutant sources, ensuring adequate ventilation, managing moisture, and using filtration and monitoring strategies.

Source Control

- Use low-emission materials with low volatile organic compounds (VOCs) and formaldehyde content to reduce chemical pollutants indoors.
- Avoid combustion appliances inside or ensure proper venting to the outdoors.
- Ban indoor smoking and limit use of harsh cleaning agents or pesticides.
- Regular cleaning and dust control techniques reduce particulate accumulation and allergens.

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Ventilation

- Provide adequate outdoor air exchange based on occupancy and room function to dilute indoor pollutants.
- Use mechanical ventilation systems with controls to maintain airflow rates and indoor air freshness.
- Employ demand-controlled ventilation linked with CO_2 sensors to optimize air exchange without energy waste.

Moisture and Humidity Control

- Maintain indoor humidity in the range of 30–60% to prevent mold growth and related microbial contaminants.
- Repair leaks and manage water ingress promptly to avoid conditions conducive to biological pollutants.

Filtration and Air Cleaning

- Incorporate high-efficiency air filters (e.g., MERV 13 or better) in HVAC systems to capture particulates.
- Use portable air purifiers in problem areas if needed, especially ones with HEPA filters.
- UV germicidal irradiation can be used within HVAC systems to reduce microbial contamination.

Indoor Air Quality Monitoring

- Implement continuous or periodic IAQ monitoring with sensors for CO_2, particulates, VOCs, and humidity.
- Use monitoring data to manage ventilation and filtration dynamically for maintaining IAQ within acceptable limits.

Design and Maintenance

- Design buildings with attention to airflow patterns that avoid pollutant accumulation zones.
- Regularly maintain HVAC and ventilation equipment for optimal performance.
- Implement green building certification standards (e.g., LEED, WELL, GRIHA, IGBC) that mandate IAQ measures and verification.

Together, these strategies help ensure acceptable IAQ levels by minimizing sources, continually renewing indoor air, managing environmental conditions, and verifying through monitoring, supporting occupant health and comfort in green buildings and other indoor environments.

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